課程目錄 — 每次上課內容

2026/09/09–2026/10/28

日期	課程主題	內容標題	頁碼
09/09	MOSFET 電流與導通參數	三極管區、飽和區電流公式；導通參數 K_n, k_n', β；氧化層電容、通道尺寸。	2
09/16	接面、通道與夾止條件	pn 接面逆偏；截止、反轉與錐形通道；夾止邊界、Overdrive；I_D–V_{DS} 特性與斜率。	3–4
09/23	通道電荷與電流推導	導通電阻；通道夾止；閘極電容與平均通道電位；電荷、載子速度、渡越時間與電流逐步推導；場依存效應。	5–6
09/30	負載、CMOS 與輸出電阻	共源級與負載線；汲源等效；二極體連接負載；CMOS 尺寸匹配、反相器與負載電容；通道長度調變、Early 電壓與 r_o。	7–10
10/07	轉導、體效應與寄生電容	g_m 等價式與權衡；通道調變；pn 逆偏與體效應 g_{mb}；小訊號模型；寄生電容、HW 10/07；共源級大訊號與 VTC。	11–20
10/14	共源級負載與源極退化	電阻負載增益、隱函數微分；MOS 等效電阻；理想電流源與本質增益；二極體 / PMOS 負載；源極退化與吸收定理。	21–28
10/21	源極退化、邏輯閘與輸出電阻	雙埠模型、G_m 與連鎖律；線性化與電阻比；10/21 小考；NAND、AOI22、德摩根公式；測試源、R_{out} 倍增與有限 r_o。	29–34
10/28	完整增益與源極隨耦器	步階測試、有限 R_D 直接推導與增益分解；理想電流源負載例題；Source Follower 大訊號、體效應、R_{out}、戴維寧與負載模型。	35–41

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2026/09/09 — MOSFET 電流與導通參數

nMOS 三極管區

$$V_{GS} > V_{TH}, \qquad 0 \le V_{DS} < V_{GS} - V_{TH}$$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[2(V_{GS} - V_{TH}) V_{DS} - V_{DS}^2 \right]$$

nMOS 飽和區

$$V_{GS} > V_{TH}, \qquad V_{DS} \ge V_{GS} - V_{TH}$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

導通參數

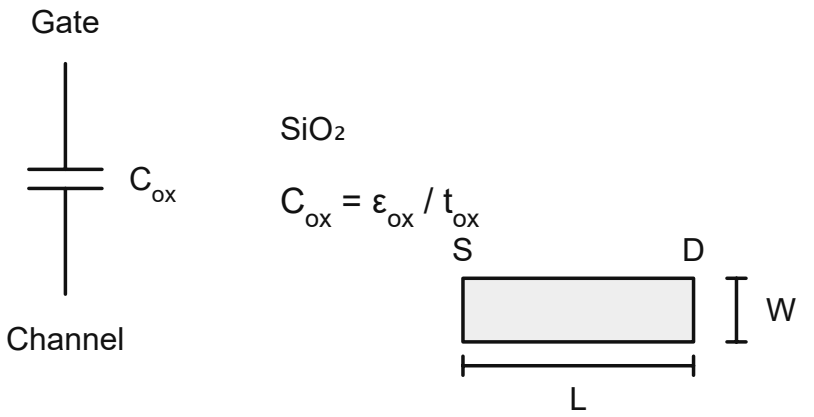
$$K_n = \frac{1}{2} \mu_n C_{ox} \frac{W}{L}$$

$$k_n' = \mu_n C_{ox}$$

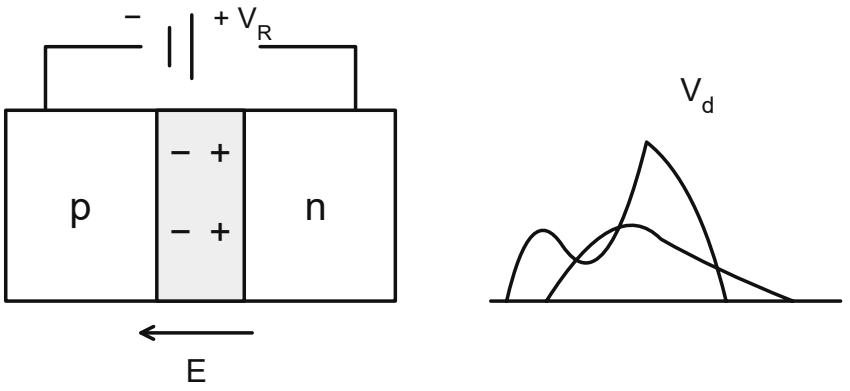
$$\beta = \mu_n C_{ox} \frac{W}{L} = k_n' \frac{W}{L} = 2K_n$$

$$\mu_n:\ \text{mobility}, \quad C_{ox}:\ \mathrm{F/m^2}, \quad K_n, k_n', \beta:\ \mathrm{A/V^2}$$

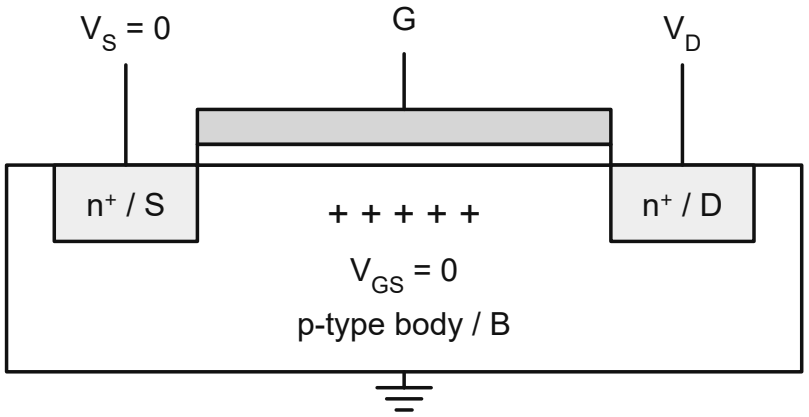
氧化層電容與通道尺寸



p–n 接面逆向偏壓

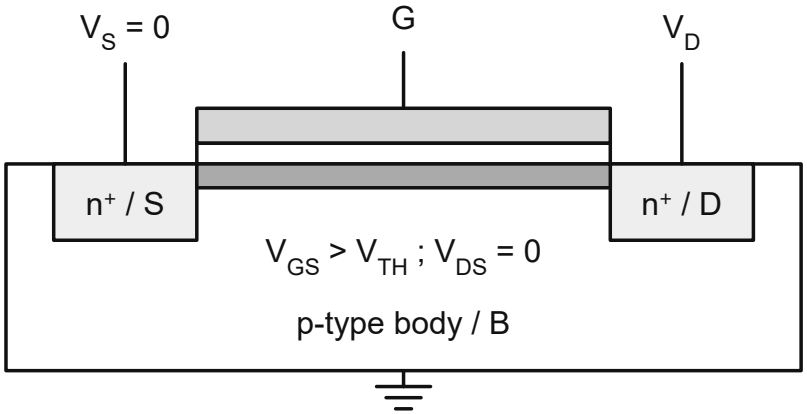


nMOS 截止

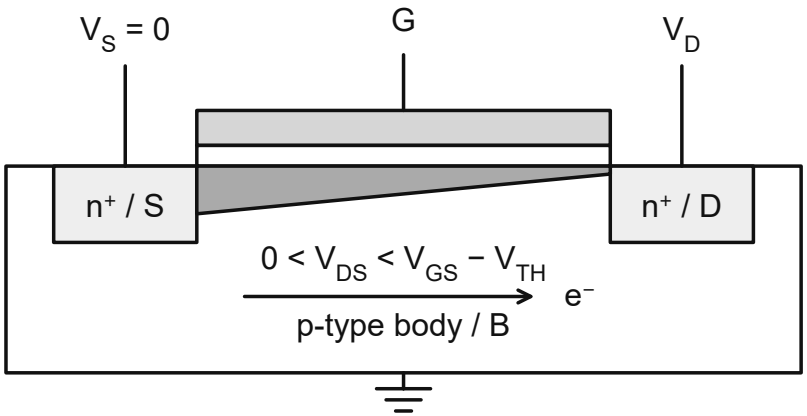


$$V_{GS} < V_{TH} \quad \Longrightarrow \quad I_D \simeq 0$$

均勻反轉通道



線性區錐形通道



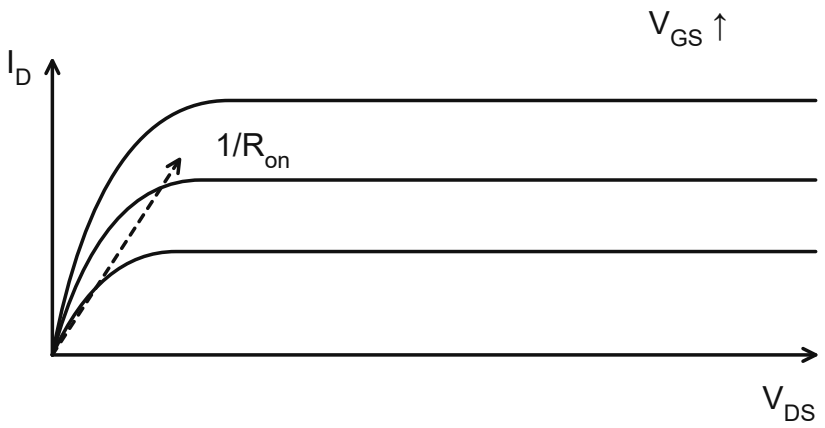
$$V_{GD} = V_{GS} - V_{DS}$$

夾止邊界與 Overdrive

$$
\begin{aligned}
V_{GD} &= V_{TH} \\
V_G - V_D &= V_{TH} \\
V_{GS} - V_{DS} &= V_{TH} \\
V_{DS,\text{sat}} &= V_{GS} - V_{TH} \equiv V_{OV}
\end{aligned}
$$

$$
\begin{aligned}
&\text{Triode :} && V_{GS} > V_{TH},\ V_{DS} < V_{OV},\ V_{GD} > V_{TH} \\
&\text{Saturation :} && V_{GS} > V_{TH},\ V_{DS} \geq V_{OV},\ V_{GD} \leq V_{TH}
\end{aligned}
$$

ID–VDS 特性與原點斜率



深線性區導通電阻

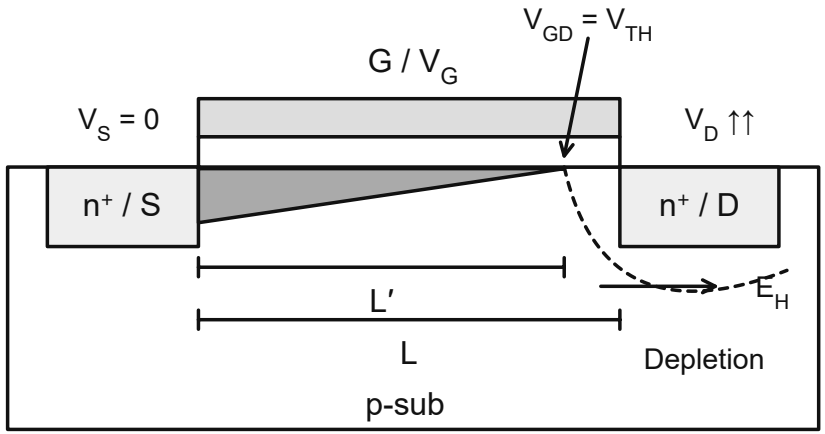
$$V_{DS} \ll 2(V_{GS} - V_{TH})$$

$$I_{DS} \simeq \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})V_{DS}$$

$$V_{DS} = R_{on} I_{DS}$$

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})}$$

通道夾止與 L′ < L



閘極電容與平均通道電位

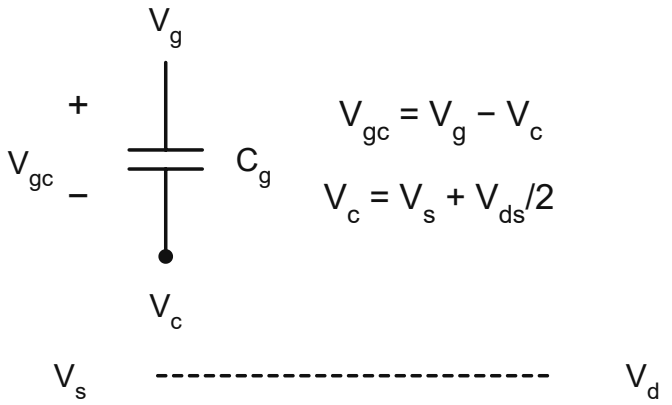
$$C = \varepsilon \frac{A}{d}, \quad A = WL, \quad d = t_{ox}$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}, \qquad C_g = C_{ox} W L$$

$$\varepsilon_{r,\mathrm{SiO_2}} \simeq 3.9$$

$$V_c = \frac{V_s + V_d}{2} = V_s + \frac{V_{ds}}{2}$$

$$V_{gc} = V_g - V_c = V_g - \left(V_s + \frac{V_{ds}}{2}\right) = V_{gs} - \frac{V_{ds}}{2}$$



通道電荷與渡越時間

$$
\begin{aligned}
Q_{\text{channel}} &= CV = C_g(V_{gc} - V_t) \\
&= C_{ox} W L \left(V_{gs} - \frac{V_{ds}}{2} - V_t \right) \\
E &= \frac{V_{ds}}{L}, \qquad v = \mu E = \mu \frac{V_{ds}}{L} \\
t &= \frac{L}{v} = \frac{L^2}{\mu V_{ds}} \\
\mu_n &\simeq 500 \sim 700 \ \text{cm}^2/(\text{V s}) \quad (E \text{ low})
\end{aligned}
$$

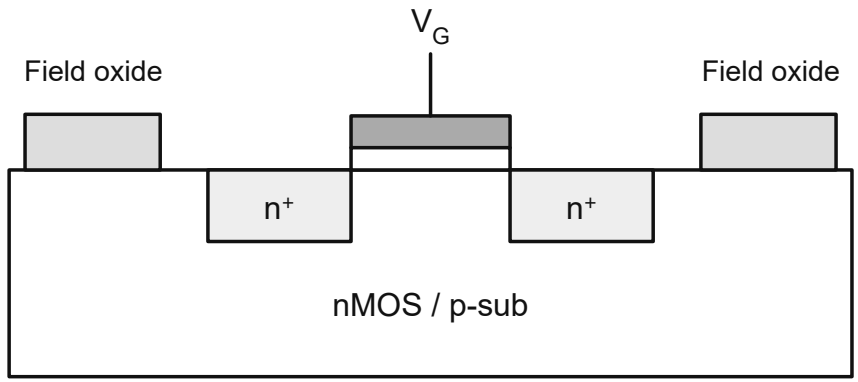
線性區電流逐步推導

$$
\begin{aligned}
I_{ds} &= \frac{Q_{\text{channel}}}{t} = \frac{C_g(V_{gc} - V_t)}{L/v} \\
&= \frac{C_{ox} W L [(V_{gs} - V_{ds}/2) - V_t]}{L/v} \\
&= C_{ox} W [(V_{gs} - V_{ds}/2) - V_t] \mu E \\
&= C_{ox} W [(V_{gs} - V_{ds}/2) - V_t] \mu \frac{V_{ds}}{L} \\
&= \mu C_{ox} \frac{W}{L} \left(V_{gs} - V_t - \frac{V_{ds}}{2} \right) V_{ds} \\
&= \frac{1}{2} \mu C_{ox} \frac{W}{L} \left[2(V_{gs} - V_t) - V_{ds} \right] V_{ds} \\
&= \beta \left(V_{gs} - V_t - \frac{V_{ds}}{2} \right) V_{ds}, \quad \beta = \mu C_{ox} \frac{W}{L}
\end{aligned}
$$

遷移率與臨界電壓相依性

$$
\mu = \mu(V_{GS}, E_{\text{vertical}})
$$

$$
V_t = f(V_d), \qquad V_t = f(L), \qquad V_t = f(V_{SB})
$$



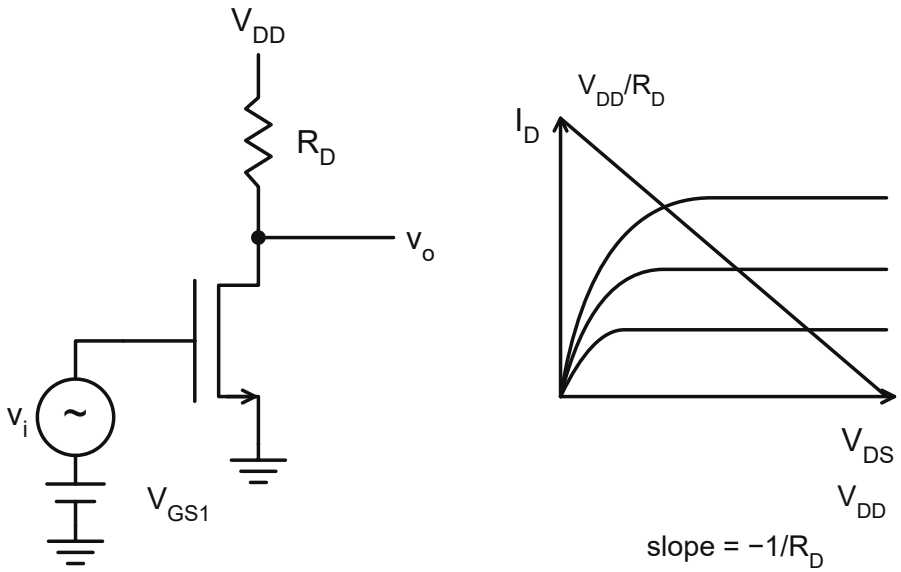
電阻負載共源極與負載線

$$v_{GS}(t) = V_{GS1} + v_i(t)$$

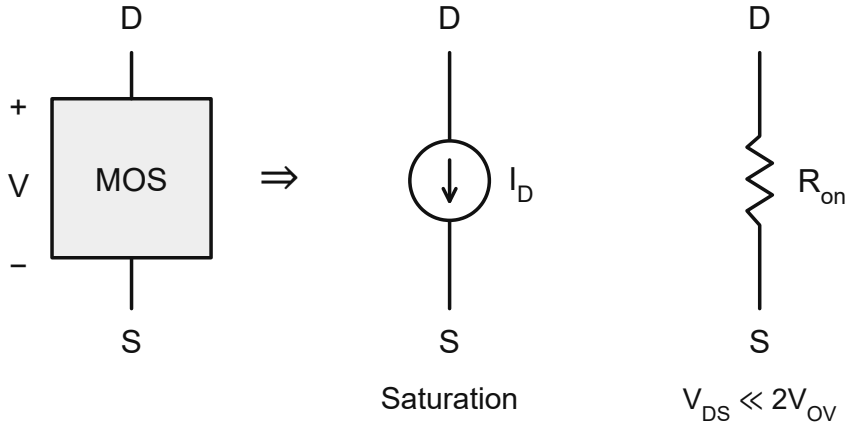
$$V_{DD} = I_D R_D + V_{DS}$$

$$I_D = \frac{V_{DD} - V_{DS}}{R_D} = -\frac{V_{DS}}{R_D} + \frac{V_{DD}}{R_D}$$

$$\frac{dI_D}{dV_{DS}} = -\frac{1}{R_D}, \qquad v_o = v_{DS}$$

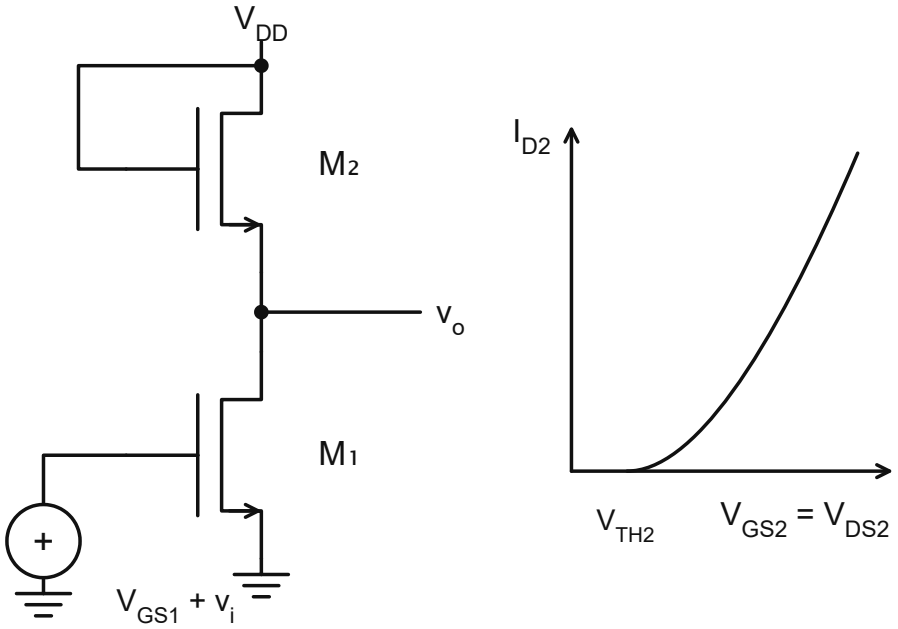


汲源埠等效



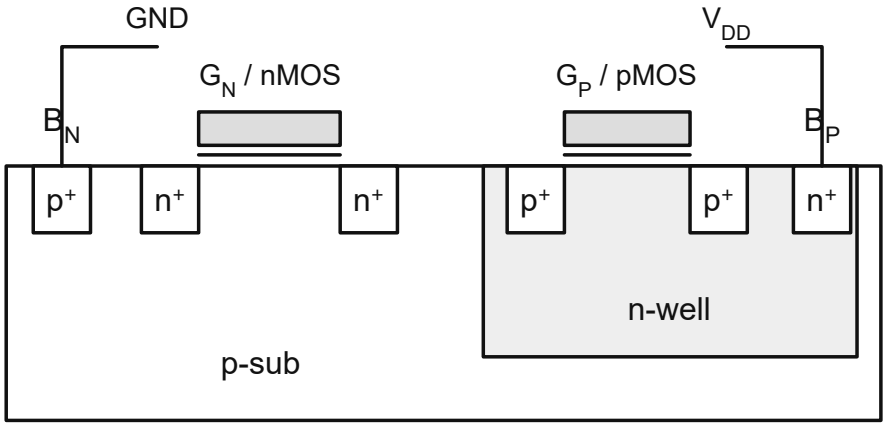
二極體連接 nMOS 負載

$$G_2 = D_2, \quad V_{GD2} = 0, \quad V_{DS2} = V_{GS2}$$
$$V_O = V_{DS1} = V_{DD} - V_{DS2}$$
$$I_{D2} = K_{n2}(V_{GS2} - V_{TH2})^2, \quad V_{GS2} > V_{TH2}$$

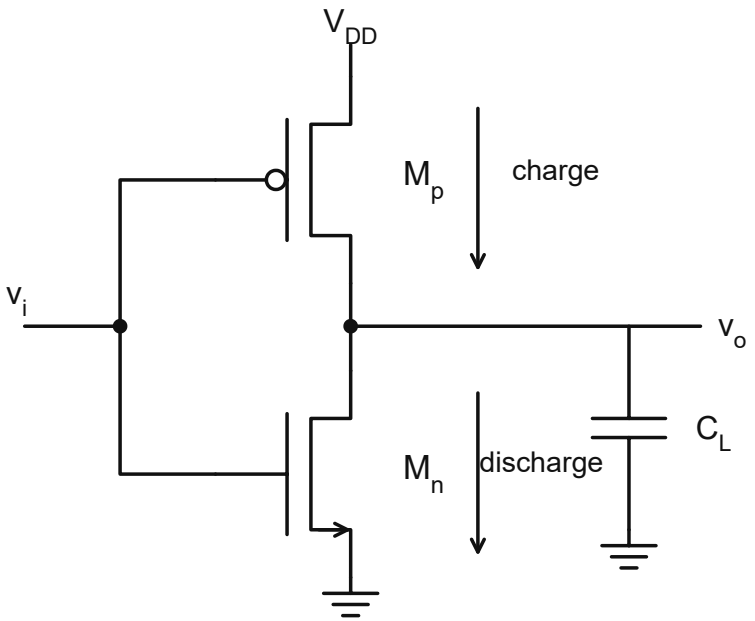


CMOS 電氣等效

$$V_{TN} = |V_{TP}|, \qquad K_n = K_p, \qquad \mu_n \neq \mu_p$$
$$K_n = \frac{1}{2}k_n'\frac{W_n}{L_n}, \quad K_p = \frac{1}{2}k_p'\frac{W_p}{L_p}$$
$$k_n'\frac{W_n}{L_n} = k_p'\frac{W_p}{L_p}$$
$$\mu_n\frac{W_n}{L_n} = \mu_p\frac{W_p}{L_p}$$
$$\frac{(W/L)_p}{(W/L)_n} = \frac{\mu_n}{\mu_p} \quad (C_{ox,n} = C_{ox,p})$$



CMOS 反相器與負載電容



場相依遷移率

$$\mu_p = f(V_{GS}), \qquad \mu = \mu(E), \qquad v = \mu(E)E$$

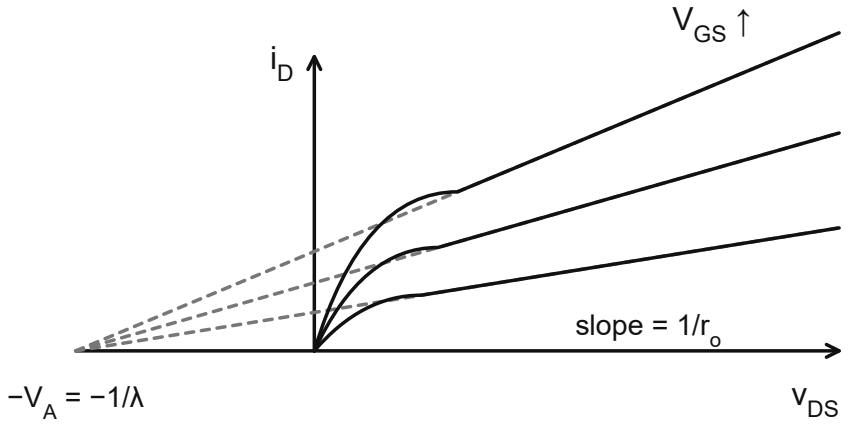
通道長度調變與 Early 電壓

$$i_D = K_n(v_{GS} - V_{TN})^2(1 + \lambda v_{DS})$$

$$\lambda > 0, \quad [\lambda] = \mathrm{V}^{-1}$$

$$1 + \lambda v_{DS} = 0 \implies i_D = 0$$

$$v_{DS} = -\frac{1}{\lambda} = -V_A, \qquad V_A = \frac{1}{\lambda}$$

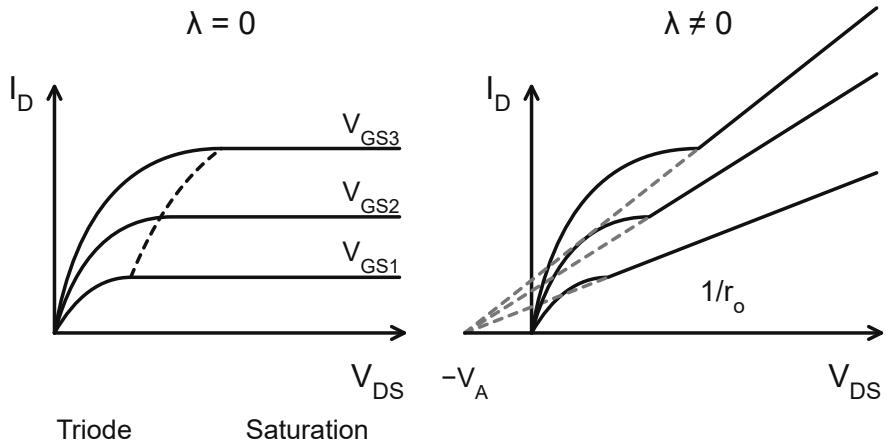


輸出電阻 ro

$$
\begin{aligned}
I_{D0} &\equiv K_n(V_{GSQ}-V_{TN})^2 \\
I_{DQ} &= I_{D0}(1+\lambda V_{DSQ}) \\
r_o &= \left(\frac{\partial i_D}{\partial v_{DS}}\right)^{-1}\Bigg|_{v_{GS}=V_{GSQ}} \\
&= \left[\lambda K_n(V_{GSQ}-V_{TN})^2\right]^{-1} \\
&= \frac{1}{\lambda I_{D0}} = \frac{V_A}{I_{D0}} \\
&= \frac{1+\lambda V_{DSQ}}{\lambda I_{DQ}} \simeq \frac{1}{\lambda I_{DQ}} = \frac{V_A}{I_{DQ}}
\end{aligned}
$$

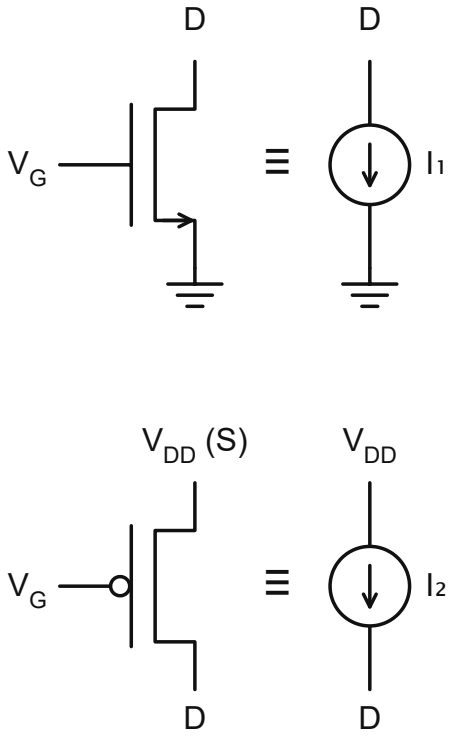
工作區域與輸出特性

$$V_{OV} = V_{GS} - V_{TH} > 0$$
$$V_{DS,\text{sat}} = V_{OV}$$
$$0 \le V_{DS} < V_{OV}: \text{ Triode}$$
$$V_{DS} \ge V_{OV}: \text{ Saturation}$$
$$V_{DS} \ll V_{OV}: \text{ Deep triode}$$



飽和區電流源模型

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2, \quad \lambda = 0$$



轉導 g_m：三個等價式

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\tfrac{W}{L}V_{OV}^2$$

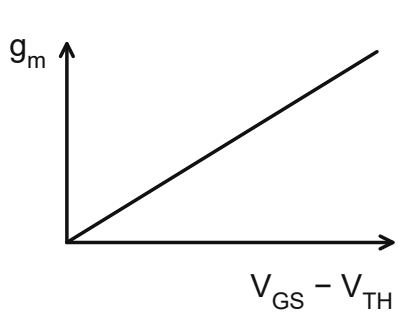
$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS},V_{BS}} = \mu_n C_{ox}\frac{W}{L}V_{OV}$$

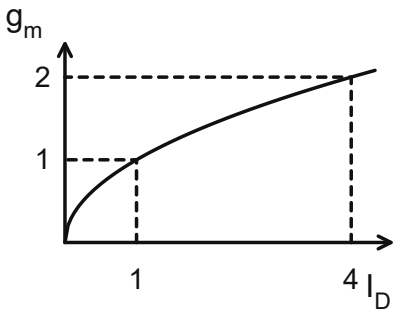
$$V_{OV} = \sqrt{\frac{2I_D}{\mu_n C_{ox}(W/L)}}$$

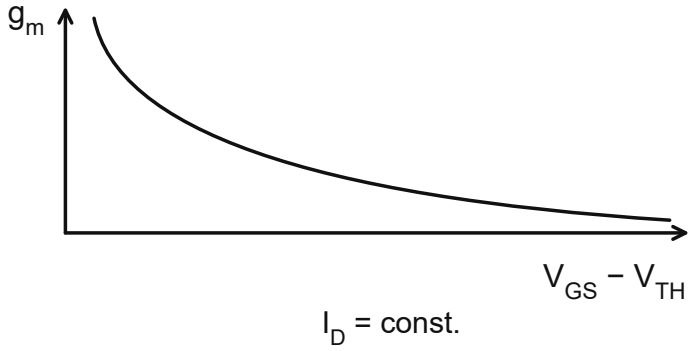
$$\boxed{g_m} = \boxed{\sqrt{2\mu_n C_{ox}\frac{W}{L}I_D} = \sqrt{4K_n I_D} = \frac{2I_D}{V_{OV}}}$$

$$K_n = \tfrac{1}{2}\mu_n C_{ox}\tfrac{W}{L}$$

轉導 g_m：三個等價式（續）







通道長度調變：泰勒展開

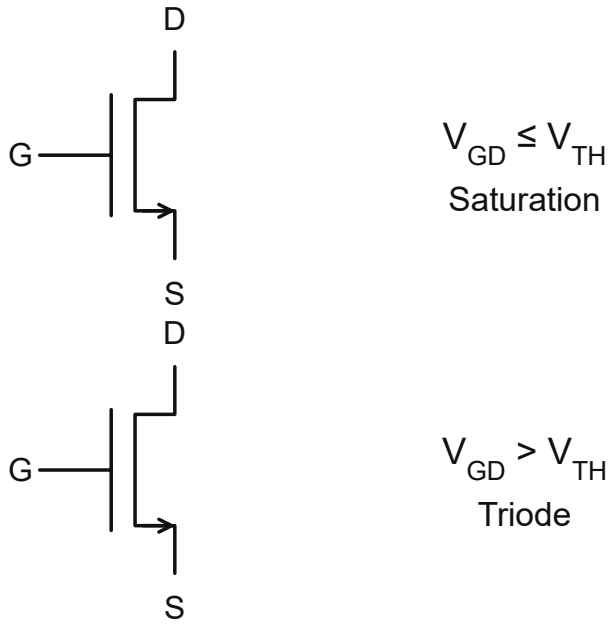
$$
\begin{aligned}
L' &= L - \Delta L, \qquad \Delta L \ll L \\
I_D &= \frac{1}{2}\mu_n C_{ox}\frac{W}{L'}V_{OV}^2 \\
\frac{1}{L'} &= \frac{1}{L-\Delta L} = \frac{1}{L}\left(1-\frac{\Delta L}{L}\right)^{-1} \\
&\simeq \frac{1}{L}\left(1+\frac{\Delta L}{L}\right) \\
\frac{\Delta L}{L} &\propto V_{DS}, \qquad \frac{\Delta L}{L} = \lambda V_{DS} \\
\boxed{I_D} &= \boxed{\frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{OV}^2(1+\lambda V_{DS})}
\end{aligned}
$$

通道調變：g_m 與 r_o

$$
\begin{aligned}
I_{D0} &= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{OV}^2 \\
I_D &= I_{D0}(1+\lambda V_{DS}) \\
g_m &= \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox}\frac{W}{L}V_{OV}(1+\lambda V_{DS}) \\
g_o &= \frac{\partial I_D}{\partial V_{DS}} = \lambda I_{D0} \\
\boxed{r_o} &= \boxed{\frac{1}{g_o} = \frac{1}{\lambda I_{D0}} = \frac{1+\lambda V_{DS}}{\lambda I_D}} \\
r_o &\simeq \frac{1}{\lambda I_D} = \frac{V_A}{I_D}, \quad V_A = \frac{1}{\lambda}
\end{aligned}
$$

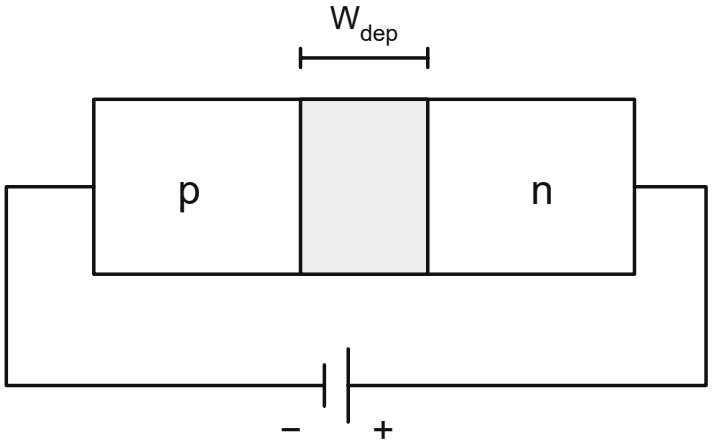
閘 - 汲跨壓判別

$$V_{DS} \gtreqless V_{GS} - V_{TH} \iff V_{GD} \lesseqgtr V_{TH}$$



pn 接面反向偏壓

$$V_R \uparrow \Rightarrow\ W_{dep} \uparrow$$

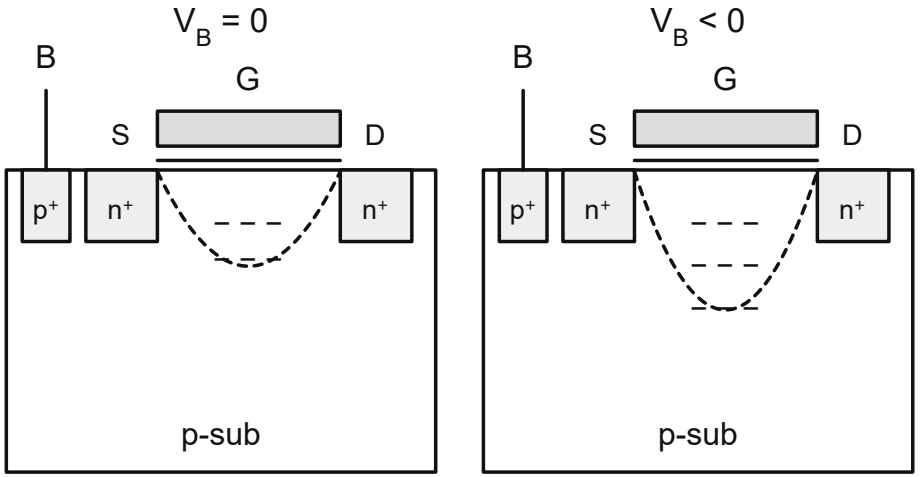


體效應：截面與臨界電壓

$$V_{SB} = V_S - V_B \geq 0, \qquad \Phi = 2|\phi_F|$$

$$\boxed{V_{TH}} = \boxed{V_{TH0} + \gamma\left(\sqrt{\Phi + V_{SB}} - \sqrt{\Phi}\right)}$$

$$\gamma \simeq 0.3\text{–}0.4\ \mathrm{V}^{1/2}, \quad V_{SB} \uparrow \Rightarrow V_{TH} \uparrow$$

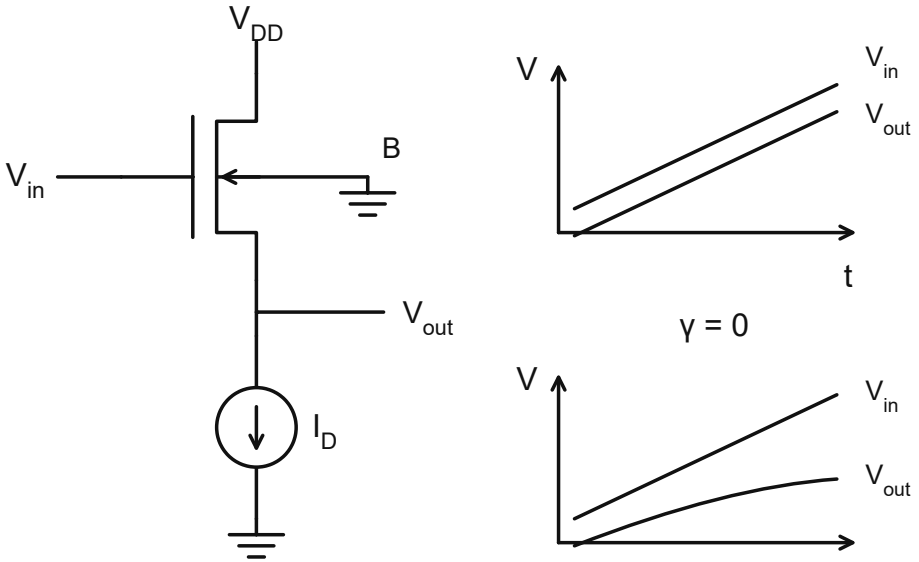


源極隨耦器與體效應

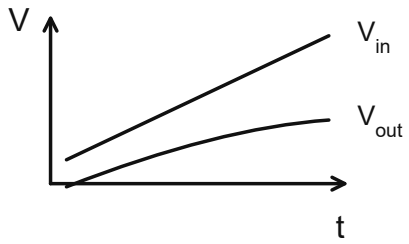
$$I_D = \tfrac{1}{2}\mu_n C_{ox}\tfrac{W}{L}(V_{in} - V_{out} - V_{TH})^2 = \text{const.}$$

$$\gamma = 0: \quad V_{in} - V_{out} = V_{TH} + V_{OV} = \text{const.}$$

$$\gamma \neq 0: \quad V_{SB} = V_{out}, \quad V_{TH} = f(V_{out})$$



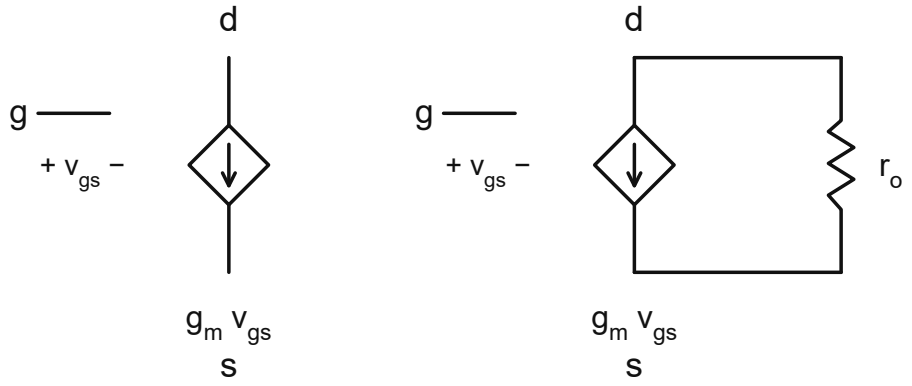
γ = 0

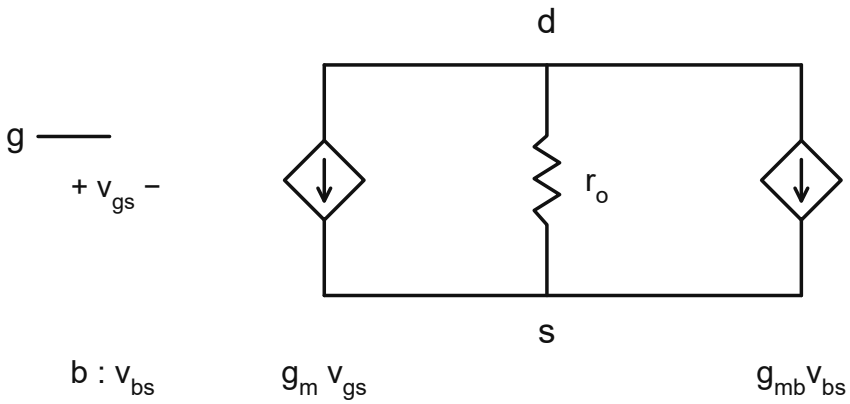


γ ≠ 0

飽和區小訊號模型

$$\boxed{i_d = g_m v_{gs} + g_{mb} v_{bs} + \frac{v_{ds}}{r_o}}, \qquad i_g = 0$$

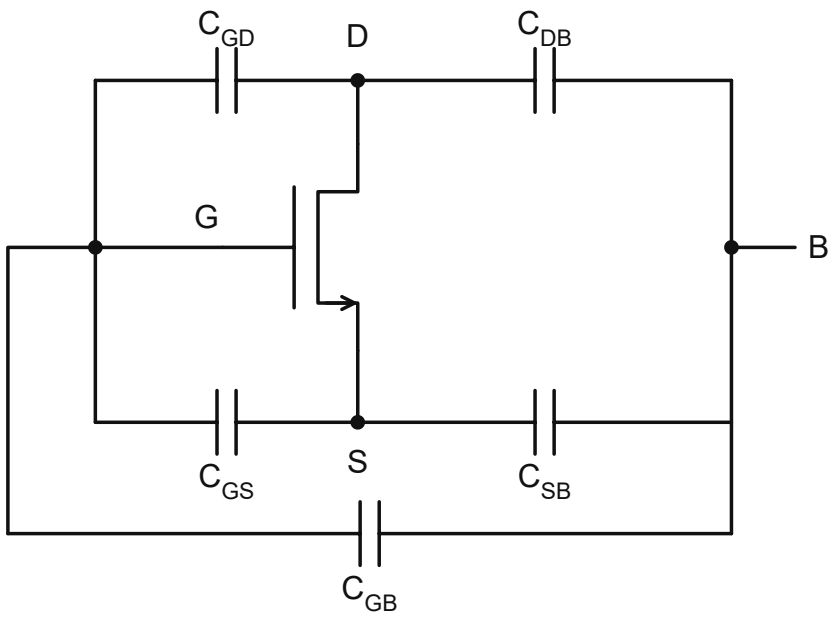




背閘轉導 g_{mb}：連鎖律

$$
\begin{aligned}
g_{mb} &\equiv \left.\frac{\partial I_D}{\partial V_{BS}}\right|_{V_{GS},V_{DS}} \\
&= \frac{\partial I_D}{\partial V_{TH}}\frac{\partial V_{TH}}{\partial V_{BS}} = -g_m\frac{\partial V_{TH}}{\partial V_{BS}} \\
V_{BS} &= -V_{SB} \\
\frac{\partial V_{TH}}{\partial V_{BS}} &= -\frac{\partial V_{TH}}{\partial V_{SB}} = -\frac{\gamma}{2\sqrt{\Phi+V_{SB}}} = -\eta \\
\boxed{g_{mb}} &= \boxed{\eta g_m}, \qquad \eta = \frac{g_{mb}}{g_m}
\end{aligned}
$$

MOSFET 五個端點寄生電容



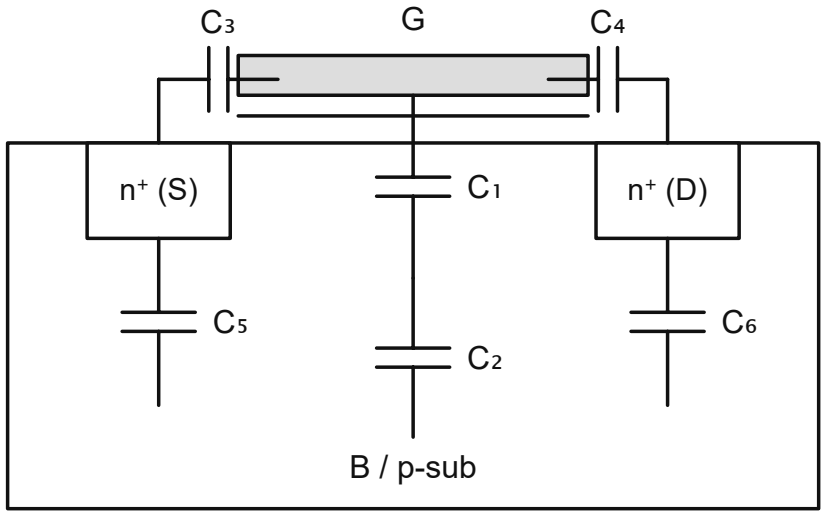
寄生電容 C_1–C_6

$$C_1 = WLC_{ox}$$

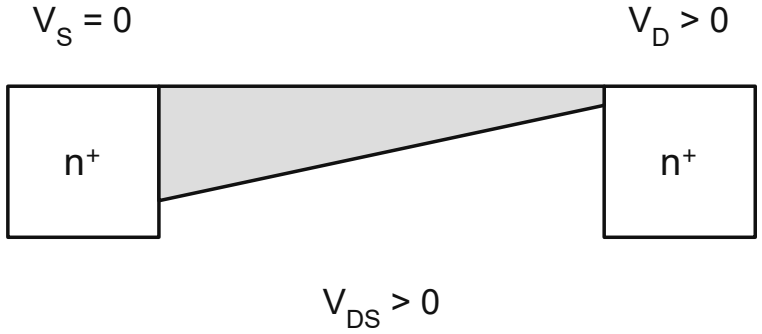
$$C_2 = C_{dep} = WL\sqrt{\frac{q\varepsilon_{si}N_{sub}}{4|\phi_F|}}$$

$$C_3 = C_4 = WL_DC_{ox} = WC_{ov}$$

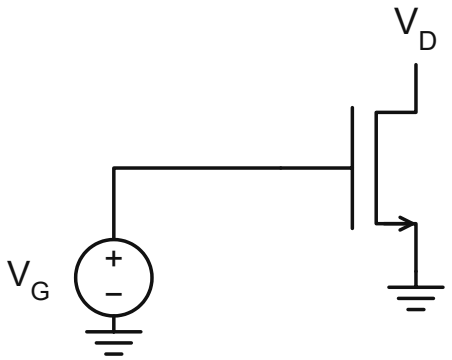
$$C_5, C_6: \quad C_j + C_{jsw}$$



偏壓下的通道形狀

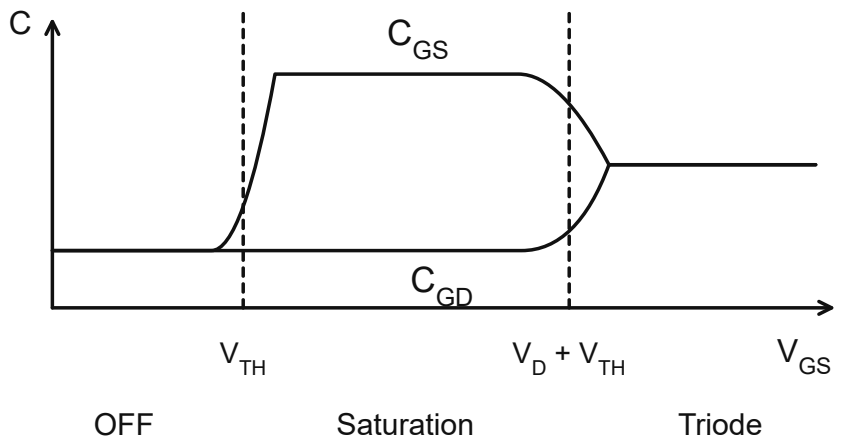


電容特性的偏壓電路



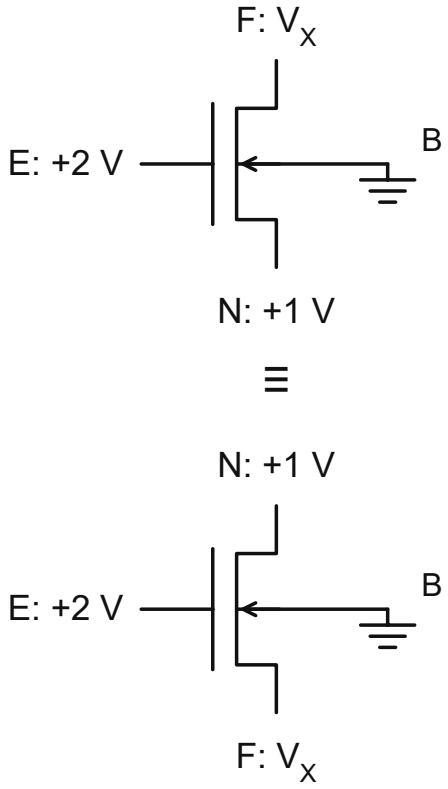
閘極電容與工作區域

	C_{GS}	C_{GD}
OFF	WC_{ov}	WC_{ov}
Saturation	$\frac{2}{3}WLC_{ox}+WC_{ov}$	WC_{ov}
Deep triode	$\frac{1}{2}WLC_{ox}+WC_{ov}$	$\frac{1}{2}WLC_{ox}+WC_{ov}$



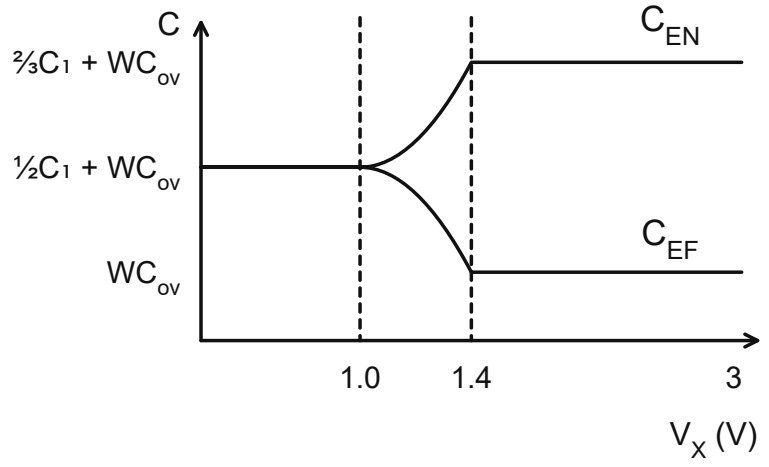
HW 2026/10/07：偏壓電路

$$V_E = 2\ \text{V},\quad V_N = 1\ \text{V},\quad V_B = 0,$$
$$V_F = V_X:\ 0 \to 3\ \text{V},\quad V_{TH} = 0.6\ \text{V},$$
$$V_E - V_{TH} = 1.4\ \text{V}$$



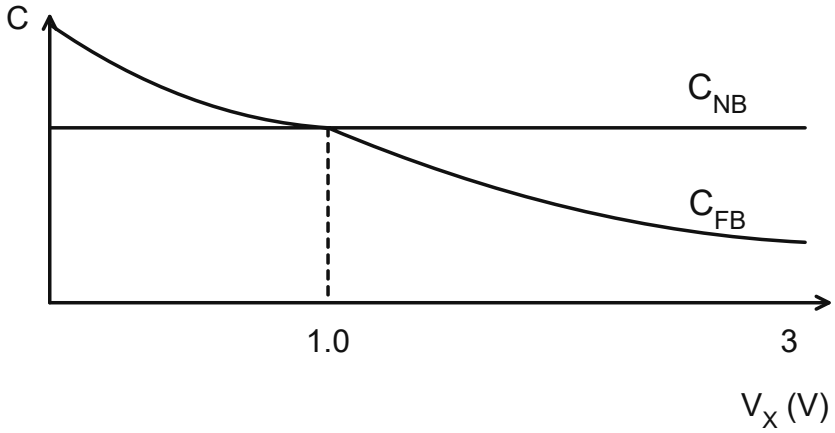
HW Fig.2：C_{EN} 與 C_{EF}

$$V_X = 1\ \text{V}: \quad C_{EN} = C_{EF} = \tfrac{1}{2}WLC_{ox} + WC_{ov}$$
$$V_X \geq 1.4\ \text{V}: \quad C_{EN} = \tfrac{2}{3}WLC_{ox} + WC_{ov}, \quad C_{EF} = WC_{ov}$$



HW Fig.3：接面電容

$$V_{NB} = 1\ \text{V}, \quad C_{NB} = \text{const.}$$
$$V_{FB} = V_X, \quad \frac{dC_{FB}}{dV_X} < 0, \quad C_{FB}(1\ \text{V}) = C_{NB}$$



電阻負載共源級：大訊號

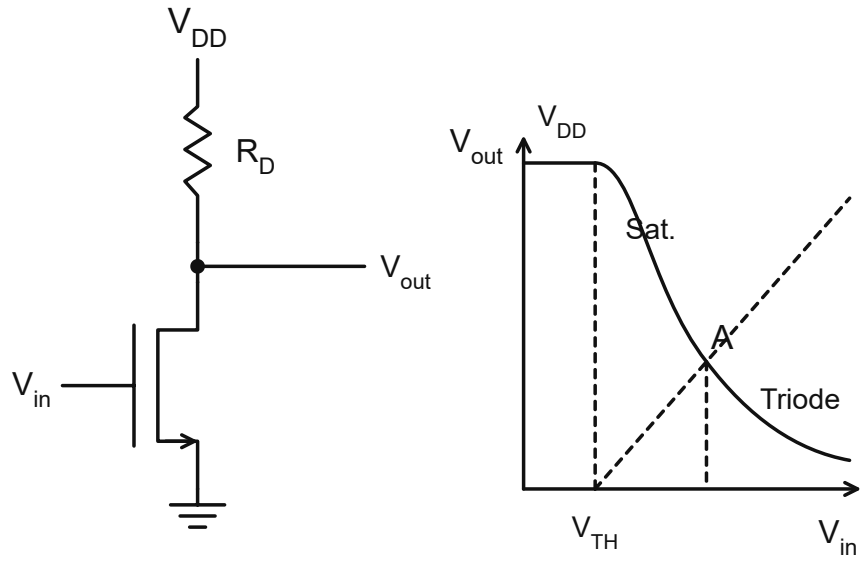
$$\lambda = 0, \qquad V_{GS} = V_{in}, \quad V_{DS} = V_{out}$$

$$V_{in} \le V_{TH}: \quad I_D = 0, \quad V_{out} = V_{DD}$$

$$V_{out} \ge V_{in} - V_{TH}: \quad V_{out} = V_{DD} - I_D R_D$$

$$= V_{DD} - \tfrac{1}{2}\mu_n C_{ox}\tfrac{W}{L}(V_{in} - V_{TH})^2 R_D$$

$$A: \quad V_{out} = V_{in} - V_{TH}$$



三極管區與深三極管區

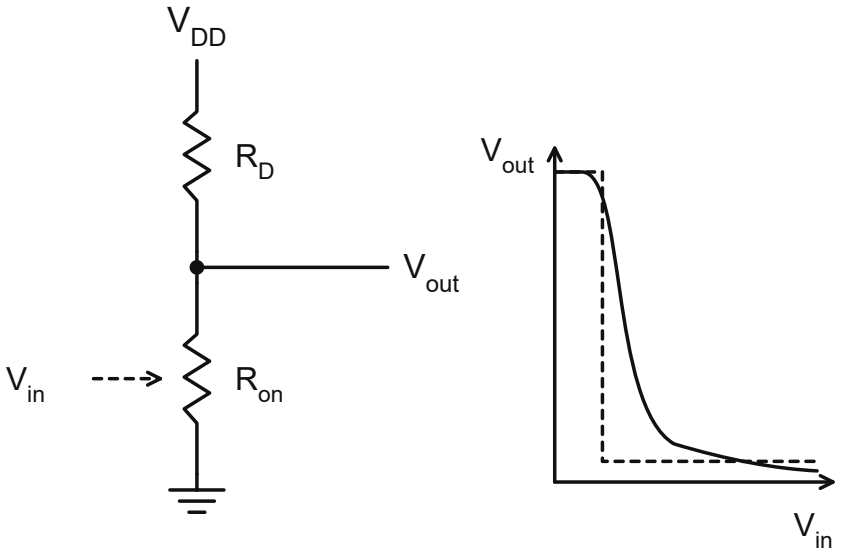
$V_{out} < V_{in} - V_{TH}$：

$$V_{out} = V_{DD} - \frac{1}{2}R_D\mu_n C_{ox}\frac{W}{L}\left[2(V_{in} - V_{TH})V_{out} - V_{out}^2\right]$$

$V_{out} \ll V_{in} - V_{TH}$：

$$R_{on} = \frac{1}{\mu_n C_{ox}(W/L)(V_{in} - V_{TH})}$$

$$\boxed{V_{out}} = \boxed{V_{DD}\frac{R_{on}}{R_D + R_{on}}}$$



2026/10/14 — 共源級負載、MOS 電阻與源極退化

電阻負載共源級：增益

$$\lambda = 0, \quad V_{GS} = V_{in}, \quad V_{DS} = V_{out}$$

$$V_{out} = V_{DD} - R_D \tfrac{1}{2}\mu_n C_{ox} \tfrac{W}{L}(V_{in} - V_{TH})^2$$

$$A_v \equiv \frac{\partial V_{out}}{\partial V_{in}} = \frac{\partial V_{DS}}{\partial V_{in}}$$

$$= -R_D \left[\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})\right]$$

$$\boxed{A_v} = \boxed{-g_m R_D}$$

增益與偏壓參數

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}, \qquad R_D = \frac{V_{RD}}{I_D}$$

$$A_v = -g_m R_D$$

$$= -\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}\, R_D$$

$$= -\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}\, \frac{V_{RD}}{I_D}$$

$$\boxed{A_v} = \boxed{-\sqrt{2\mu_n C_{ox} \frac{W}{L}}\, \frac{V_{RD}}{\sqrt{I_D}}}$$

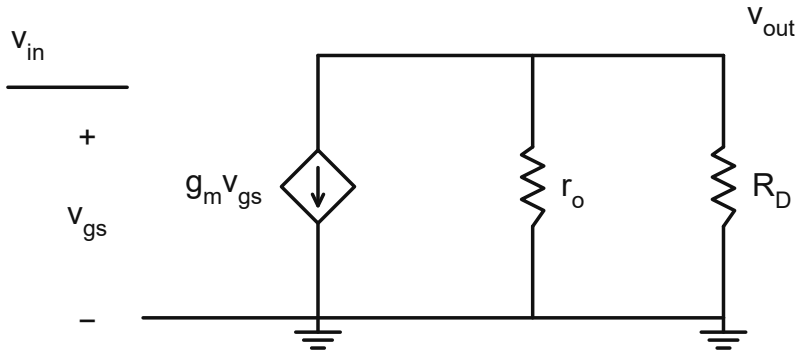
$$V_{out} = V_{DD} - V_{RD} \geq V_{in} - V_{TH}$$

共源級小訊號等效電路

$$v_{gs} = v_{in}$$

$$v_{out} = -g_m v_{gs}(R_D \parallel r_o)$$

$$\boxed{A_v} = \boxed{\frac{v_{out}}{v_{in}} = -g_m(R_D \parallel r_o)}$$

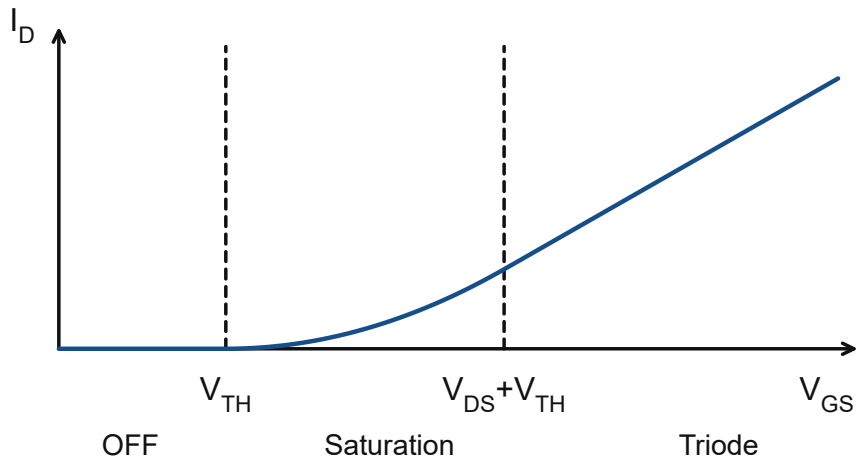


含 λ：隱函數微分

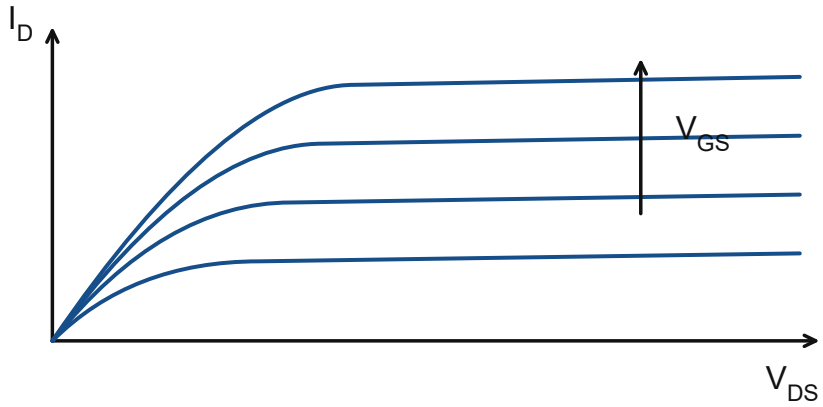
$$
\begin{aligned}
I_{D0} &= \tfrac{1}{2}\mu_n C_{ox}\tfrac{W}{L}(V_{in}-V_{TH})^2 \\
V_{out} &= V_{DD} - R_D I_{D0}(1+\lambda V_{out}) \\
A_v &= -R_D\left[\mu_n C_{ox}\tfrac{W}{L}(V_{in}-V_{TH})(1+\lambda V_{out}) + I_{D0}\lambda A_v\right] \\
&= -R_D(g_m + g_o A_v), \quad g_o = \lambda I_{D0} = \frac{1}{r_o} \\
(1+R_D g_o)A_v &= -R_D g_m \\
\boxed{A_v} &= \boxed{-\frac{R_D g_m}{1+R_D/r_o} = -g_m\frac{R_D r_o}{R_D+r_o}}
\end{aligned}
$$

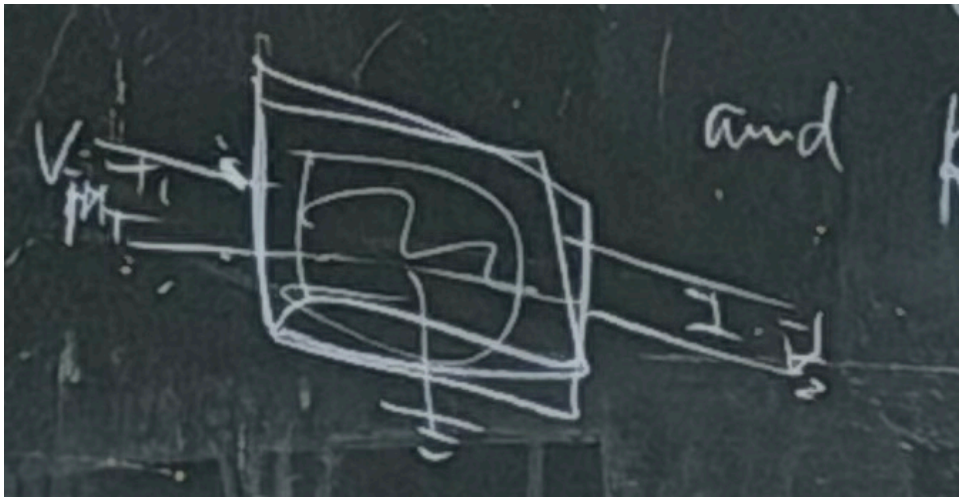
固定 V_{DS} 的轉移特性

$$
\begin{aligned}
\lambda = 0: &\quad V_{DS} = \text{const.} \\
V_{TH} < V_{GS} \le V_{DS}+V_{TH}: &\quad I_D = \tfrac{1}{2}\mu_n C_{ox}\tfrac{W}{L}(V_{GS}-V_{TH})^2 \\
V_{GS} > V_{DS}+V_{TH}: &\quad I_D = \mu_n C_{ox}\tfrac{W}{L}\left[(V_{GS}-V_{TH})V_{DS} - \tfrac{1}{2}V_{DS}^2\right]
\end{aligned}
$$

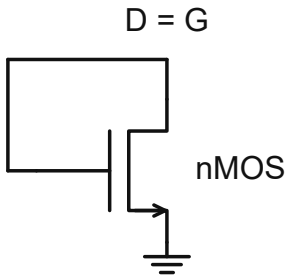


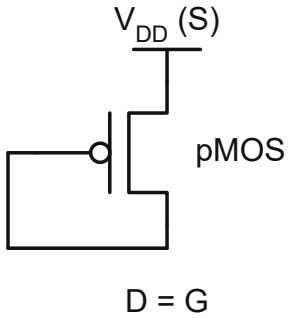
特性關係草圖（可辨線條）





二極體連接：nMOS 與 pMOS





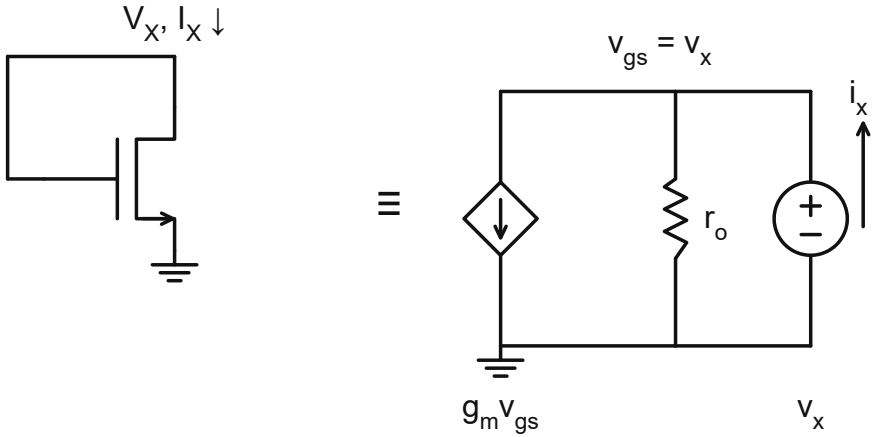
MOS 電阻：閘汲短接

$$\lambda \neq 0, \quad \gamma = 0, \quad v_1 = v_{gs} = v_x$$

$$i_x = g_m v_1 + \frac{v_x}{r_o} = \left(g_m + \frac{1}{r_o}\right) v_x$$

$$\boxed{R} = \boxed{\frac{v_x}{i_x} = \frac{1}{g_m + 1/r_o} = \frac{1}{g_m} \parallel r_o}$$

$$g_m \gg 1/r_o: \quad R \simeq \frac{1}{g_m}$$



MOS 電阻：源端輸入與體效應

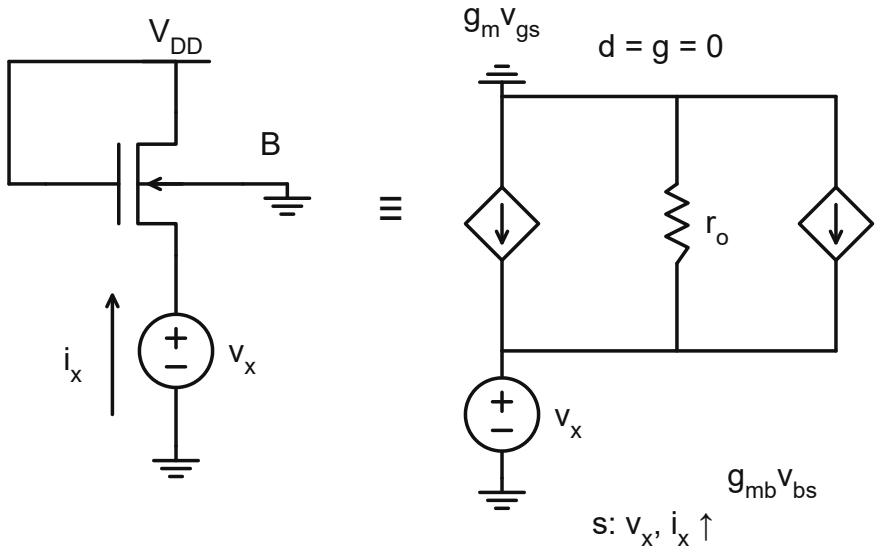
$$\lambda \neq 0, \quad \gamma \neq 0, \quad v_1 = v_{gs} = -v_x, \quad v_{bs} = -v_x$$

$$i_x = -g_m v_1 - g_{mb} v_{bs} + \frac{v_x}{r_o}$$

$$= v_x \left(g_m + g_{mb} + \frac{1}{r_o}\right)$$

$$\boxed{R} = \boxed{\frac{1}{g_m + g_{mb} + 1/r_o}}$$

$$= \frac{1}{g_m} \parallel \frac{1}{g_{mb}} \parallel r_o \simeq \frac{1}{g_m + g_{mb}}$$



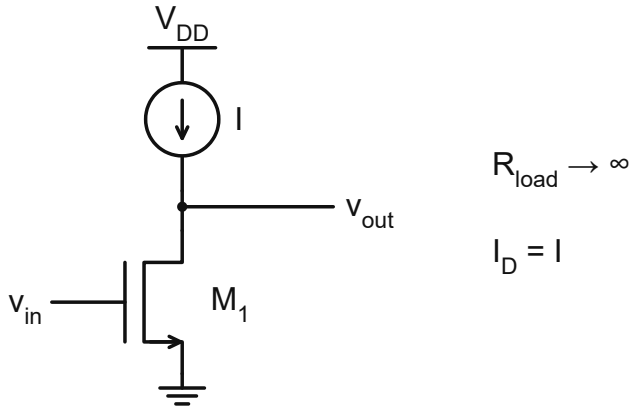
理想電流源負載與本質增益

$$A_v = -g_m(R_D \parallel r_o)$$

$$R_D \to \infty: \quad \boxed{A_v = -g_m r_o}$$

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\tfrac{W}{L}(V_{in}-V_{TH})^2(1+\lambda V_{out}) = I$$

$$dI_D = g_m\, dV_{in} + g_o\, dV_{out} = 0$$



$R_{load} \to \infty$

$I_D = I$

二極體連接負載共源級

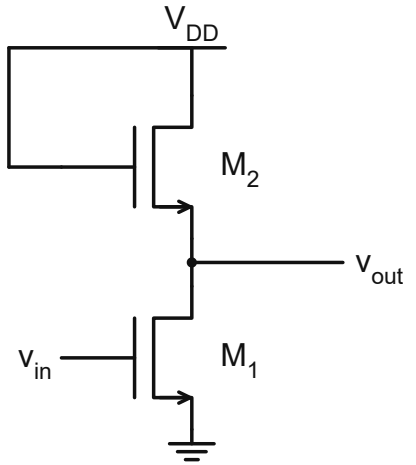
$$R_L = \frac{1}{g_{m2}+g_{mb2}}, \quad \eta_2 = \frac{g_{mb2}}{g_{m2}}$$

$$A_v = -g_{m1}R_L = -\frac{g_{m1}}{g_{m2}}\frac{1}{1+\eta_2}$$

$$= -\frac{\sqrt{2\mu_n C_{ox}(W/L)_1 I_{D1}}}{\sqrt{2\mu_n C_{ox}(W/L)_2 I_{D2}}}\frac{1}{1+\eta_2}$$

$$\boxed{A_v} = \boxed{-\sqrt{\frac{(W/L)_1}{(W/L)_2}}\frac{1}{1+\eta_2}}$$

$$\eta_2 = 0: \quad A_v = -\frac{g_{m1}}{g_{m2}}$$



$I_{D1} = I_{D2}$

$\lambda_1 = \lambda_2 = 0$



$1/(g_{m2}+g_{mb2})$

二極體負載：大訊號等電流

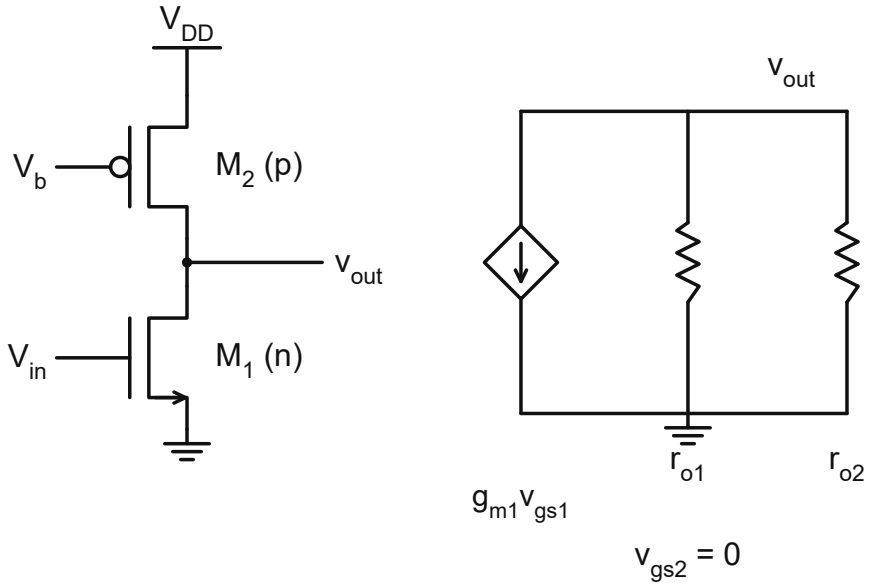
$$I_{D1} = I_{D2}, \quad \lambda_1 = \lambda_2 = 0$$
$$\tfrac{1}{2}\mu_n C_{ox}(W/L)_1(V_{in} - V_{TH1})^2 = \tfrac{1}{2}\mu_n C_{ox}(W/L)_2(V_{DD} - V_{out} - V_{TH2})^2$$
$$\sqrt{(W/L)_1}(V_{in} - V_{TH1}) = \sqrt{(W/L)_2}(V_{DD} - V_{out} - V_{TH2})$$
$$V_{in} - V_{TH1} > 0, \quad V_{DD} - V_{out} - V_{TH2} > 0$$

二極體負載：連鎖微分

$$\frac{\partial V_{TH1}}{\partial V_{in}} = 0, \quad V_{SB2} = V_{out}, \quad \frac{\partial V_{TH2}}{\partial V_{out}} = \eta_2$$
$$\sqrt{(W/L)_1}\left(1 - \frac{\partial V_{TH1}}{\partial V_{in}}\right) = \sqrt{(W/L)_2}\left[-\frac{\partial V_{out}}{\partial V_{in}} - \frac{\partial V_{TH2}}{\partial V_{out}}\frac{\partial V_{out}}{\partial V_{in}}\right]$$
$$\sqrt{(W/L)_1} = -\sqrt{(W/L)_2}(1+\eta_2)\frac{\partial V_{out}}{\partial V_{in}}$$
$$\boxed{A_v} = \boxed{-\sqrt{\frac{(W/L)_1}{(W/L)_2}}\frac{1}{1+\eta_2}}$$

PMOS 電流源負載共源級

$$M_1, M_2: \quad \text{Saturation}$$
$$v_{gs2} = 0 \quad \Rightarrow \quad g_{m2}v_{gs2} = 0$$
$$\boxed{A_v} = \boxed{-g_{m1}(r_{o1} \parallel r_{o2})}$$
$$V_{SD2,\min} = V_{SG2} - |V_{THp}|$$
$$V_{in} - V_{THn} \le V_{out} \le V_{DD} - (V_{SG2} - |V_{THp}|)$$



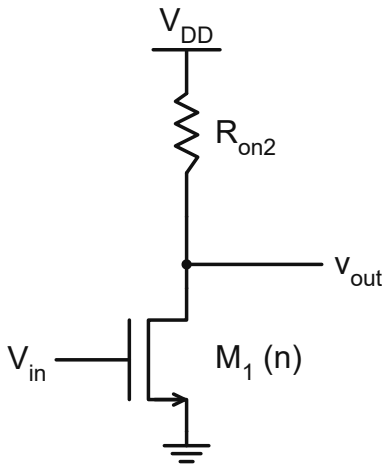
PMOS 負載進入深三極管區

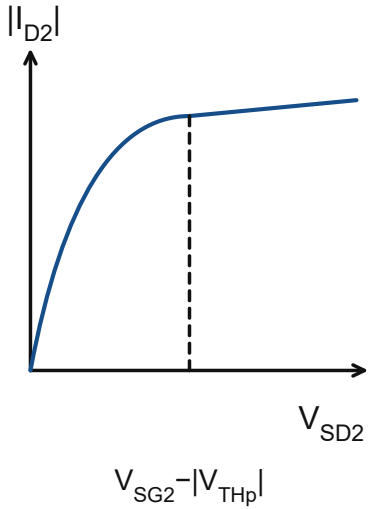
$$V_{SD2} \ll V_{SG2} - |V_{THp}|$$

$$R_{on2} = \frac{1}{\mu_p C_{ox}(W/L)_2(V_{SG2} - |V_{THp}|)}$$

$$\boxed{A_v} \simeq \boxed{-g_{m1}R_{on2}}$$

$$V_{out,\max} \to V_{DD} \quad (I_D \to 0)$$





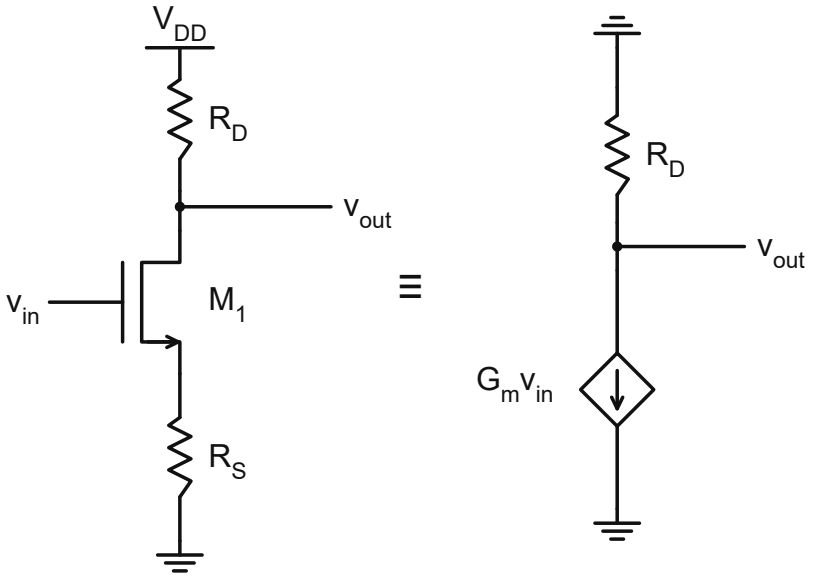
源極退化共源級與等效轉導

$$\lambda = 0, \quad \gamma = 0, \quad I_D = f(V_{GS})$$

$$G_m \equiv \frac{\partial I_D}{\partial V_{in}}, \quad g_m = \frac{\partial I_D}{\partial V_{GS}}$$

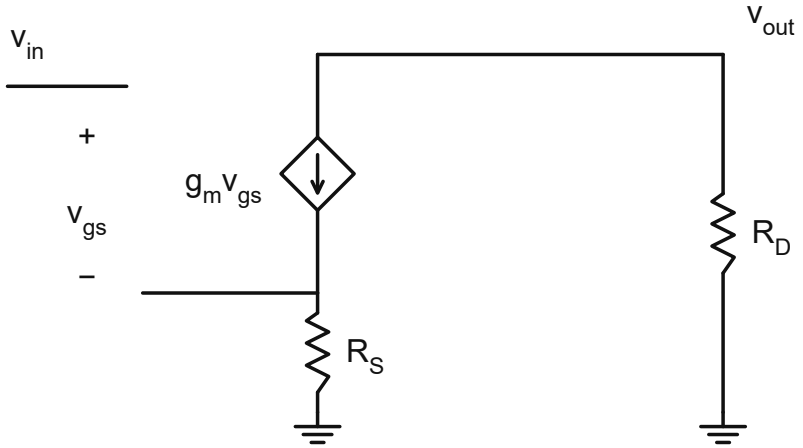
$$V_{GS} = V_{in} - I_D R_S$$

$$\frac{\partial V_{GS}}{\partial V_{in}} = 1 - R_S \frac{\partial I_D}{\partial V_{in}} = 1 - R_S G_m$$



源極退化的小訊號電路

$$
\begin{aligned}
v_s &= i_d R_S, \quad v_{gs} = v_{in} - v_s, \quad i_d = g_m v_{gs} \\
v_{out} &= -i_d R_D \\
\frac{\partial V_{out}}{\partial V_{in}} &= -\frac{\partial I_D}{\partial V_{in}} R_D = -\frac{\partial f}{\partial V_{in}} R_D \\
&= -\frac{\partial f}{\partial V_{GS}} \frac{\partial V_{GS}}{\partial V_{in}} R_D = -g_m \frac{\partial V_{GS}}{\partial V_{in}} R_D
\end{aligned}
$$

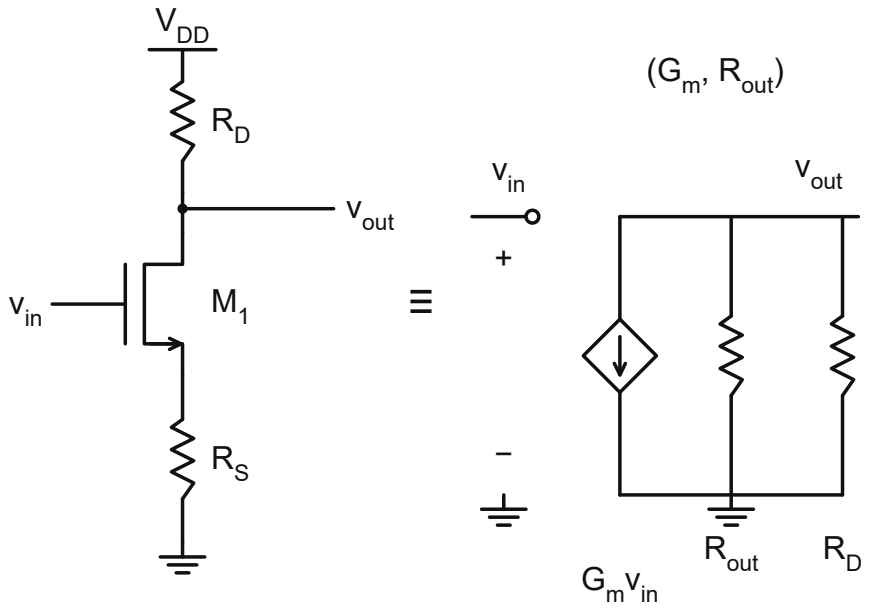


源極退化：連鎖律與吸收定理

$$
\begin{aligned}
G_m &= \frac{\partial I_D}{\partial V_{GS}} \frac{\partial V_{GS}}{\partial V_{in}} = g_m (1 - R_S G_m) \\
G_m (1 + g_m R_S) &= g_m \\
\boxed{G_m} &= \boxed{\frac{g_m}{1 + g_m R_S} = \frac{1}{1/g_m + R_S}} \\
\frac{\partial V_{out}}{\partial V_{in}} &= -\frac{\partial I_D}{\partial V_{in}} R_D = -G_m R_D \\
\boxed{A_v} &= \boxed{-\frac{g_m R_D}{1 + g_m R_S} = -\frac{R_D}{1/g_m + R_S}}
\end{aligned}
$$

$$
\frac{1}{g_m} \ll R_S : \quad G_m \simeq \frac{1}{R_S}, \quad A_v \simeq -\frac{R_D}{R_S}
$$

源極退化 CS 與雙埠模型



$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2, \quad I_D = f(V_{GS}), \quad V_{in} \neq V_{GS}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}}, \qquad G_m = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0}, \qquad R_{out} = \left.\frac{v_x}{i_x}\right|_{v_{in}=0}$$

$$i_{out} = G_m v_{in}, \qquad A_v = \frac{\partial V_{out}}{\partial V_{in}} = -G_m(R_{out} \parallel R_D)$$

連鎖律：$\lambda = \gamma = 0$

$$V_{out} = V_{DD} - I_D R_D, \qquad V_{GS} = V_{in} - I_D R_S$$

$$A_v = -\frac{\partial I_D}{\partial V_{in}} R_D = -\frac{\partial I_D}{\partial V_{GS}}\frac{\partial V_{GS}}{\partial V_{in}} R_D$$

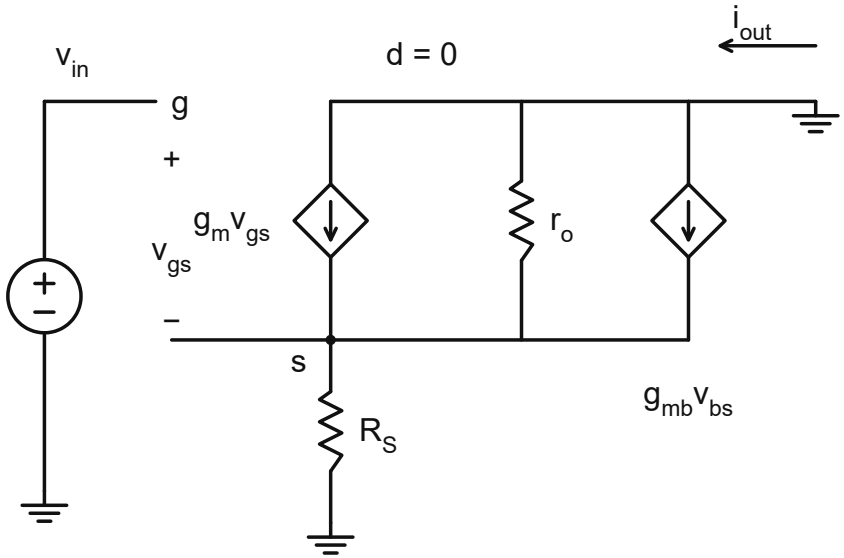
$$\frac{\partial V_{GS}}{\partial V_{in}} = 1 - \frac{\partial I_D}{\partial V_{in}} R_S = 1 - G_m R_S$$

$$G_m = g_m(1 - G_m R_S) \Rightarrow G_m(1 + g_m R_S) = g_m \Rightarrow \boxed{G_m = \frac{g_m}{1 + g_m R_S}}$$

$$A_v = -g_m\left(1 - \frac{g_m R_S}{1 + g_m R_S}\right) R_D$$

$$= -g_m \frac{R_D}{1 + g_m R_S} = -G_m R_D$$

短路轉導：$r_o,\ g_{mb}$



$$v_d = 0, \quad v_x = v_{RS} = i_{out}R_S, \quad v_1 = v_{in} - i_{out}R_S, \quad v_{bs} = -v_x = -i_{out}R_S$$

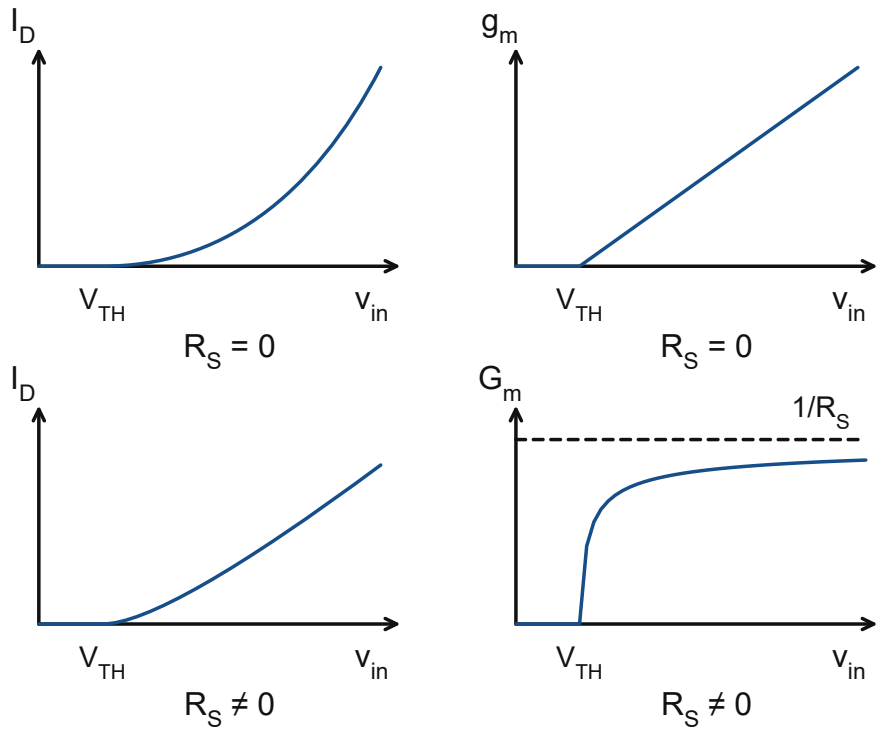
$$i_{r_o} = \frac{0 - v_x}{r_o} = -\frac{i_{out}R_S}{r_o}$$

$$\begin{aligned} i_{out} &= g_m v_1 + g_{mb}v_{bs} + i_{r_o} \\ &= g_m(v_{in} - i_{out}R_S) - g_{mb}i_{out}R_S - \frac{i_{out}R_S}{r_o} \end{aligned}$$

$$i_{out}\left(1 + g_m R_S + g_{mb}R_S + \frac{R_S}{r_o}\right) = g_m v_{in}$$

$$\boxed{G_m = \frac{g_m}{1 + (g_m + g_{mb})R_S + R_S/r_o} = \frac{g_m r_o}{R_S + r_o + (g_m + g_{mb})R_S r_o}}$$

線性化與電阻比



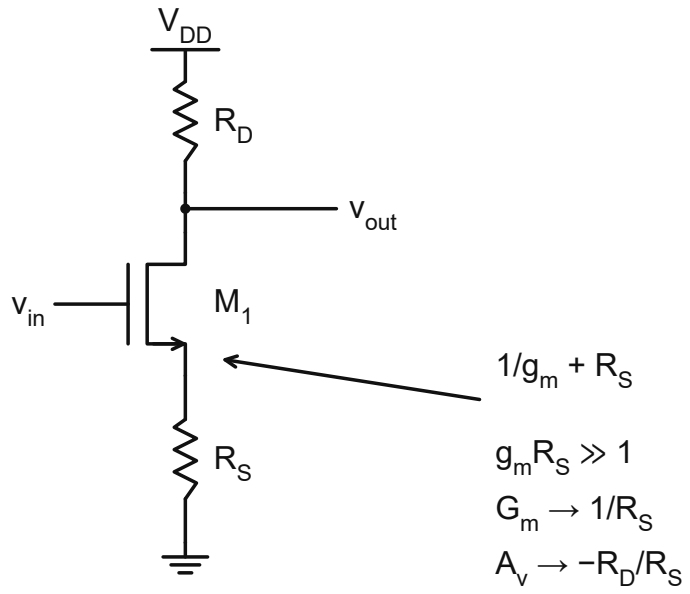
$$\lambda = \gamma = 0: \quad G_m = \frac{g_m}{1 + g_m R_S} = \frac{1}{1/g_m + R_S}, \qquad A_v = -\frac{R_D}{1/g_m + R_S}$$

$$R_S = 0: \quad G_m = g_m, \quad A_v = -g_m R_D$$

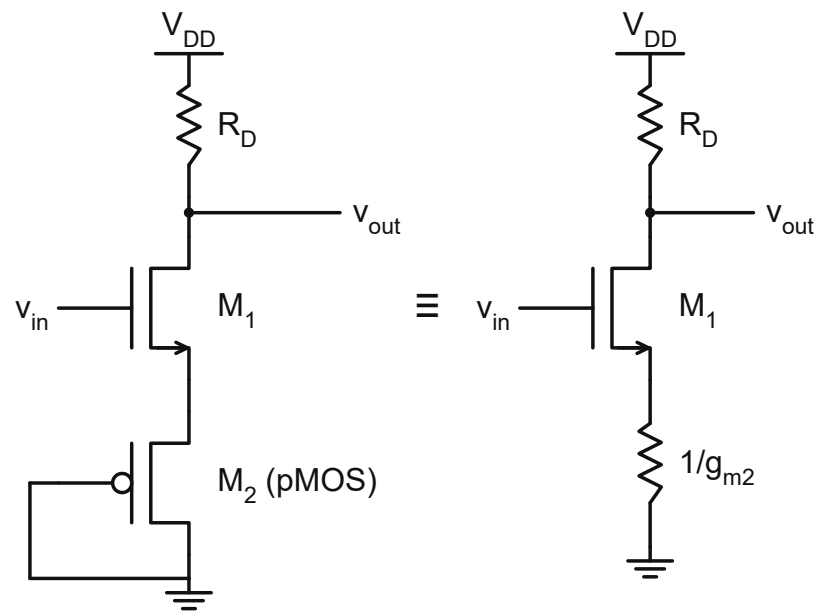
$$1/g_m \gg R_S: \quad G_m \approx g_m$$

$$g_m R_S \gg 1: \quad G_m \approx \frac{1}{R_S}, \quad A_v \approx -\frac{R_D}{R_S}$$

源極路徑等效電阻



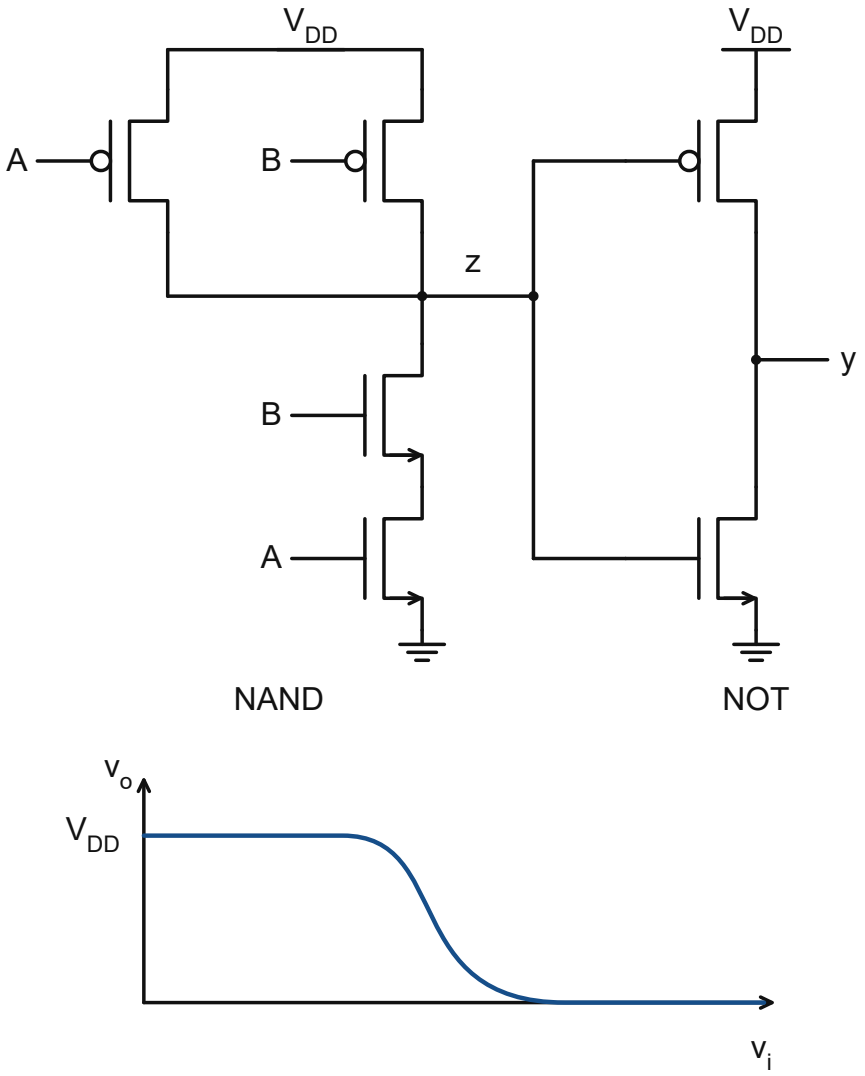
10/21 小考：$A_v = ?$



$$M_1:\text{NMOS},\quad M_2:\text{PMOS},\quad \lambda=0,\quad \gamma=0$$

$$R_S=\frac{v_x}{i_x}=\frac{1}{g_{m2}},\qquad \boxed{A_v=-\frac{R_D}{1/g_{m1}+1/g_{m2}}=-\frac{g_{m1}g_{m2}}{g_{m1}+g_{m2}}R_D}$$

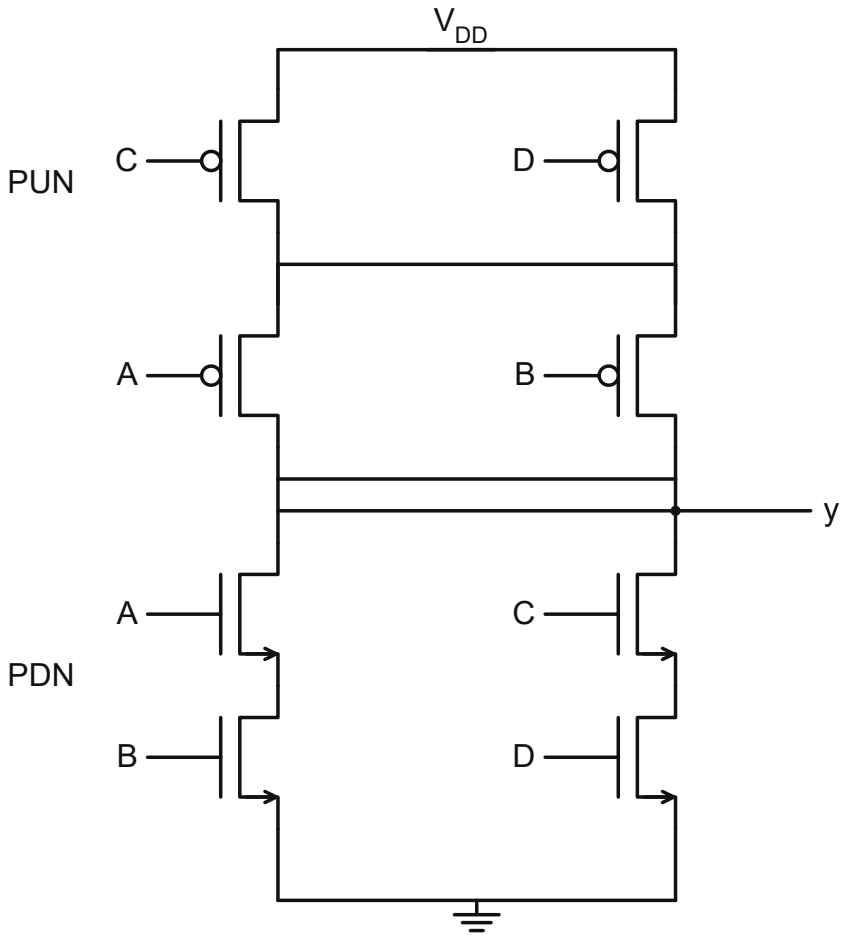
CMOS NAND 與反相器



CMOS NAND 與反相器（續）

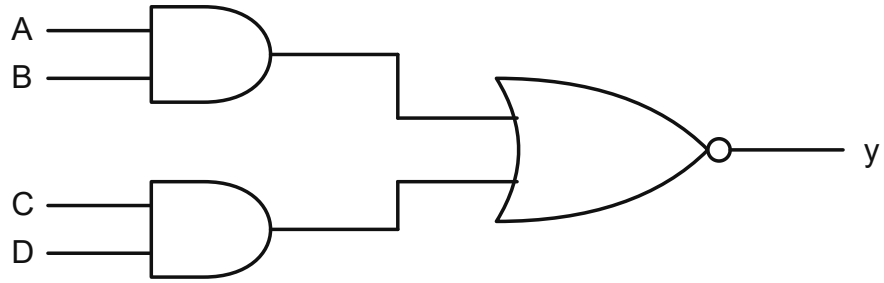
$$z = \overline{AB}, \qquad y = \overline{z} = AB$$

CMOS 複合邏輯閘 AOI22

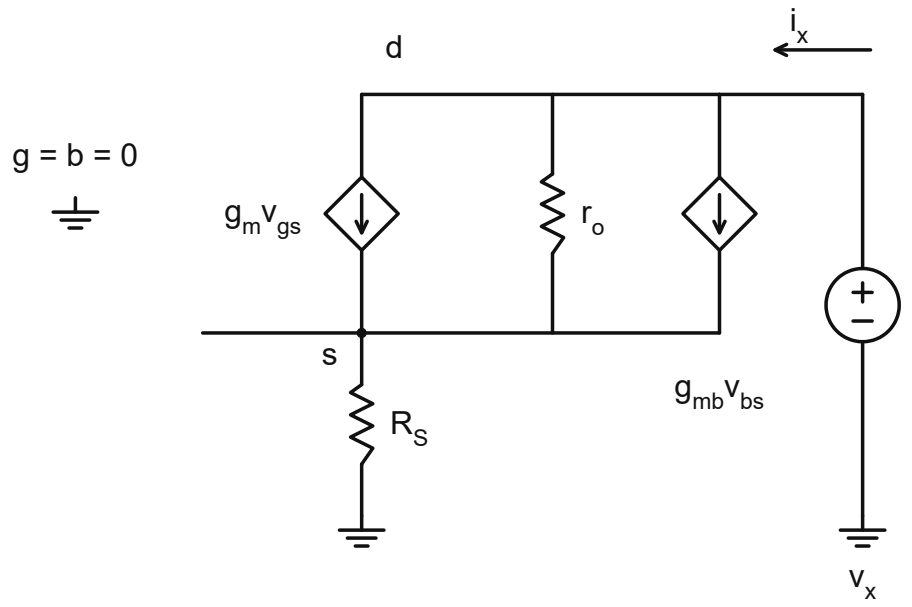


$$\overline{y} = AB + CD, \qquad \boxed{y = \overline{AB + CD} = (\overline{A} + \overline{B})(\overline{C} + \overline{D})}$$

AND–OR–INVERT 邏輯圖



輸出電阻：測試源與 KCL



$$v_{in} = v_g = v_b = 0, \qquad v_{RS} = i_x R_S, \qquad v_1 = v_{gs} = v_{bs} = -i_x R_S$$

$$i_x = g_m v_{gs} + i_{r_o} + g_{mb} v_{bs}$$

$$i_{r_o} = i_x - (g_m + g_{mb}) v_1 = i_x + (g_m + g_{mb}) R_S i_x$$

$$\begin{aligned} v_x &= v_{RS} + v_{r_o} = i_x R_S + i_{r_o} r_o \\ &= i_x R_S + [i_x + (g_m + g_{mb}) R_S i_x] r_o \\ &= i_x [R_S + r_o + (g_m + g_{mb}) R_S r_o] \end{aligned}$$

輸出電阻倍增

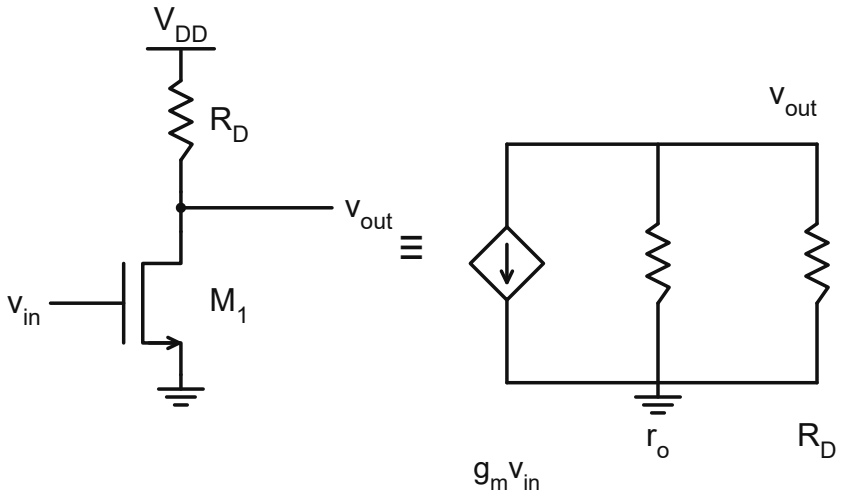
$$\boxed{R_{out} = \frac{v_x}{i_x}} = R_S + r_o + (g_m + g_{mb}) r_o R_S$$

$$= [1 + (g_m + g_{mb}) r_o] R_S + r_o$$

$$= [1 + (g_m + g_{mb}) R_S] r_o + R_S$$

$(g_m + g_{mb}) r_o \gg 1$: $\boxed{R_{out} \approx r_o + (g_m + g_{mb}) r_o R_S = [1 + (g_m + g_{mb}) R_S] r_o}$

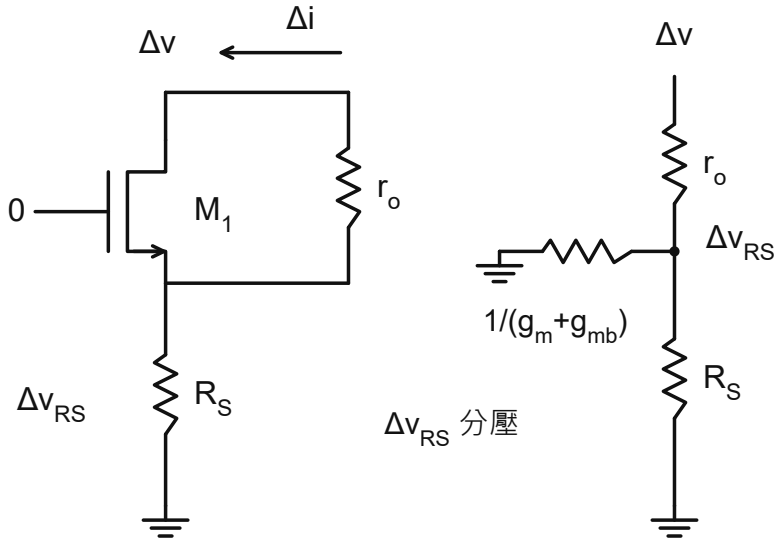
接地源極 CS：有限 r_o



$R_S = 0$: $v_{out} = -g_m v_{in} (r_o \parallel R_D)$, $\boxed{A_v = -g_m (r_o \parallel R_D)}$

$r_o \to \infty$: $A_v = -g_m R_D$; $R_D \to \infty$: $A_v = -g_m r_o$

步階測試與源極等效電阻



$R_S = 0: \quad \Delta v_{RS} = 0; \qquad v_g = v_b = 0, \qquad R_p = \frac{1}{g_m} \parallel \frac{1}{g_{mb}} \parallel R_S = \frac{1}{g_m + g_{mb} + 1/R_S}$

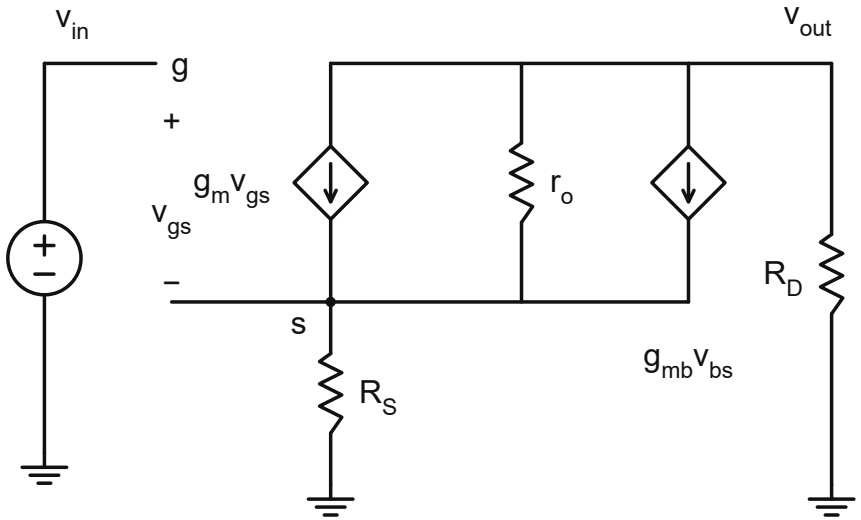
$$\Delta v_{RS} = \Delta v \frac{R_p}{r_o + R_p}$$

$$\begin{aligned} \Delta i = \frac{\Delta v_{RS}}{R_S} &= \frac{\Delta v}{R_S} \frac{1/(g_m + g_{mb} + 1/R_S)}{r_o + 1/(g_m + g_{mb} + 1/R_S)} \\ &= \frac{\Delta v}{R_S[1 + r_o(g_m + g_{mb} + 1/R_S)]} \\ &= \frac{\Delta v}{R_S + r_o + (g_m + g_{mb}) r_o R_S} \end{aligned}$$

步階測試與源極等效電阻（續）

$$\boxed{\frac{\Delta v}{\Delta i} = R_S + r_o + (g_m + g_{mb}) r_o R_S = R_{out}}$$

有限 R_D：直接法推導 A_v



$$i_{RS} = -\frac{v_{out}}{R_D}, \quad v_{RS} = -\frac{v_{out}R_S}{R_D}, \quad v_1 = v_{in} + \frac{v_{out}R_S}{R_D}, \quad v_{bs} = \frac{v_{out}R_S}{R_D}$$

$$i_{r_o} = i_{RS} - (g_m v_1 + g_{mb} v_{bs}) = -\frac{v_{out}}{R_D} - g_m\left(v_{in} + \frac{v_{out}R_S}{R_D}\right) - g_{mb}\frac{v_{out}R_S}{R_D}$$

$$v_{out} = i_{r_o} r_o + v_{RS} = -\frac{v_{out}r_o}{R_D} - \left[g_m\left(v_{in} + \frac{v_{out}R_S}{R_D}\right) + g_{mb}\frac{v_{out}R_S}{R_D}\right] r_o - \frac{v_{out}R_S}{R_D}$$

$$v_{out}\left[1 + \frac{r_o}{R_D} + \frac{g_m r_o R_S}{R_D} + \frac{g_{mb} r_o R_S}{R_D} + \frac{R_S}{R_D}\right] = -g_m r_o v_{in}$$

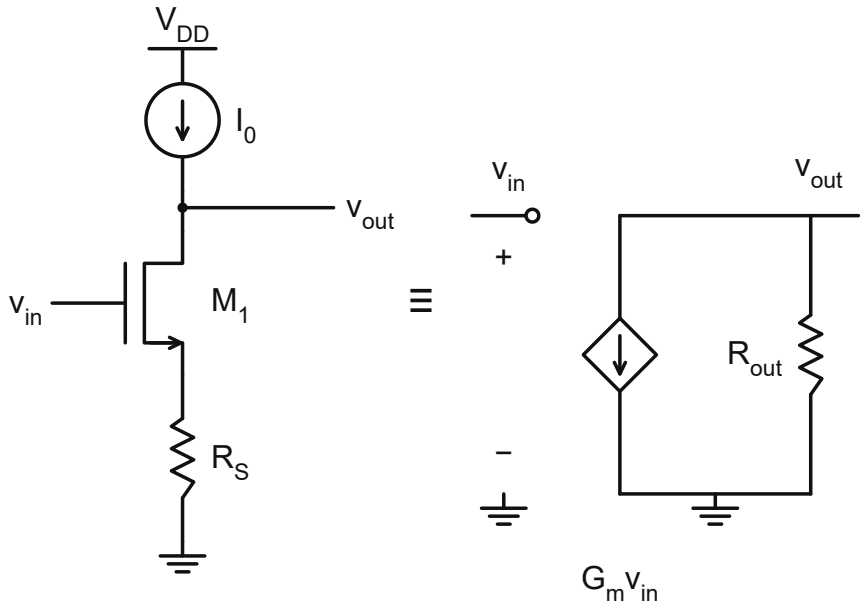
$$\boxed{A_v = -\frac{g_m r_o R_D}{R_D + R_S + r_o + (g_m + g_{mb}) r_o R_S}}$$

增益分解

$$D = R_S + r_o + (g_m + g_{mb}) r_o R_S = R_{out}, \qquad \frac{A}{B} = \frac{A}{C}\frac{C}{B}$$

$$\begin{aligned} A_v &= -\frac{g_m r_o R_D}{R_D + D} \\ &= -\frac{g_m r_o}{D}\frac{D R_D}{R_D + D} \\ &= \boxed{-G_m (R_D \parallel R_{out})} \end{aligned}$$

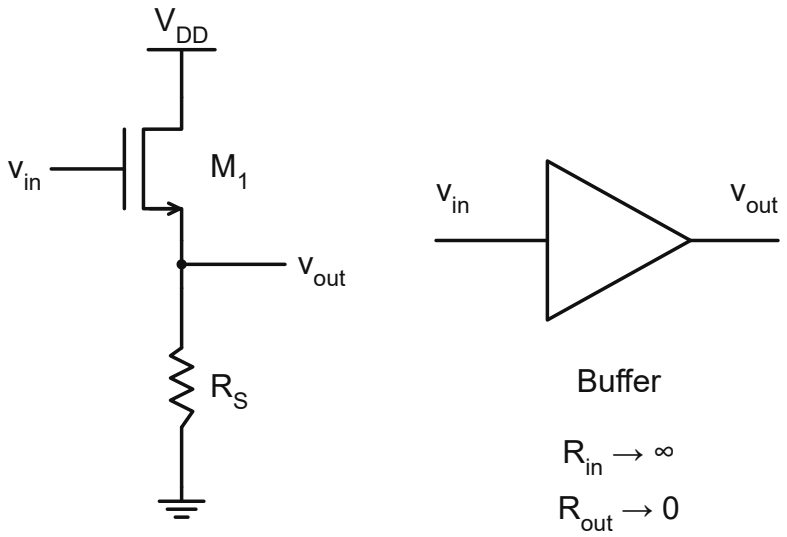
例題：理想電流源負載 CS，$A_v = ?$



$$r_{I_0} = \infty, \quad i_{RS} = 0, \quad v_{RS} = 0, \quad R_D \to \infty$$

$$\begin{aligned} A_v &= -G_m R_{out} \\ &= -\frac{g_m r_o}{R_S + r_o + (g_m + g_{mb}) R_S r_o} [R_S + r_o + (g_m + g_{mb}) R_S r_o] \\ &= \boxed{-g_m r_o} \end{aligned}$$

Source Follower / Common Drain / Buffer



$$\lambda = 0, \qquad V_{GS} = V_{in} - V_{out}, \quad V_{SB} = V_{out}$$

$$V_{in} \le V_{TH0}: \quad I_D = 0, \quad V_{out} = 0$$

$$V_{in} > V_{TH0}, \quad V_{DD} \ge V_{in} - V_{TH}: \quad \boxed{\frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{in} - V_{out} - V_{TH})^2 R_S = V_{out}}$$

體效應與大訊號微分

$$A_v=\frac{\partial V_{out}}{\partial V_{in}},\quad \eta=\frac{\partial V_{TH}}{\partial V_{SB}}=\frac{\gamma}{2\sqrt{2\phi_F+V_{out}}}=\frac{g_{mb}}{g_m}$$

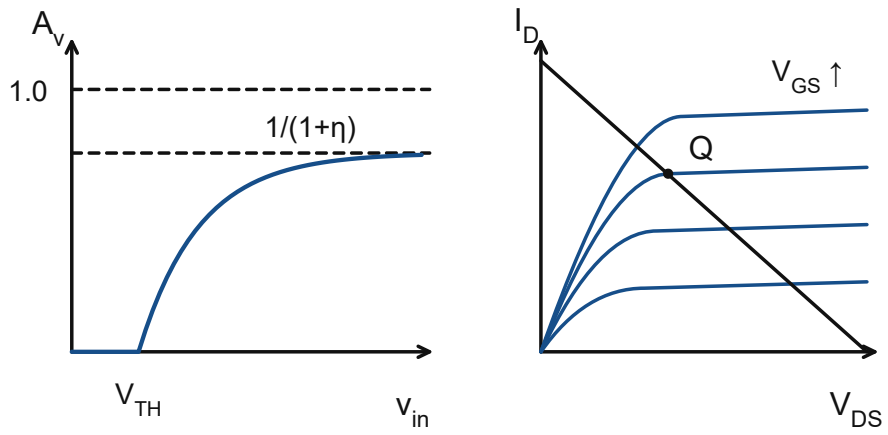
$$\frac{\partial V_{TH}}{\partial V_{in}}=\frac{\partial V_{TH}}{\partial V_{SB}}\frac{\partial V_{SB}}{\partial V_{in}}=\frac{\partial V_{TH}}{\partial V_{out}}\frac{\partial V_{out}}{\partial V_{in}}=\eta A_v$$

$$\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{out}-V_{TH})(1-A_v-\eta A_v)R_S=A_v$$

$$g_mR_S(1-A_v-\eta A_v)=A_v\Rightarrow g_mR_S=A_v[1+(1+\eta)g_mR_S]$$

$$\boxed{A_v=\frac{g_mR_S}{1+(1+\eta)g_mR_S}=\frac{g_mR_S}{1+(g_m+g_{mb})R_S}=\frac{g_m}{1/R_S+g_m+g_{mb}}}$$

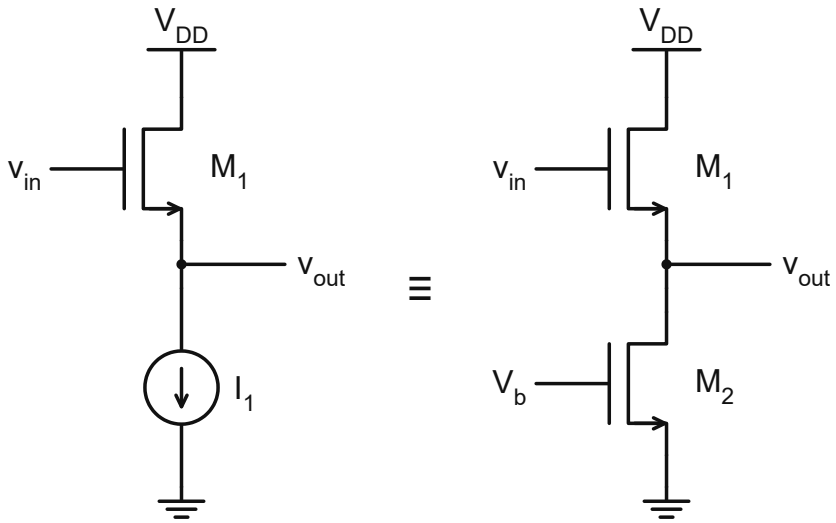
增益曲線與平方律



$$(g_m+g_{mb})R_S\gg 1:\quad A_v\approx\frac{g_m}{g_m+g_{mb}}=\frac{1}{1+\eta},\qquad \eta\approx 0.2$$

$$I_D=k(V_{GS}-V_{TH})^2,\qquad 1.5^2=2.25,\quad 2^2=4$$

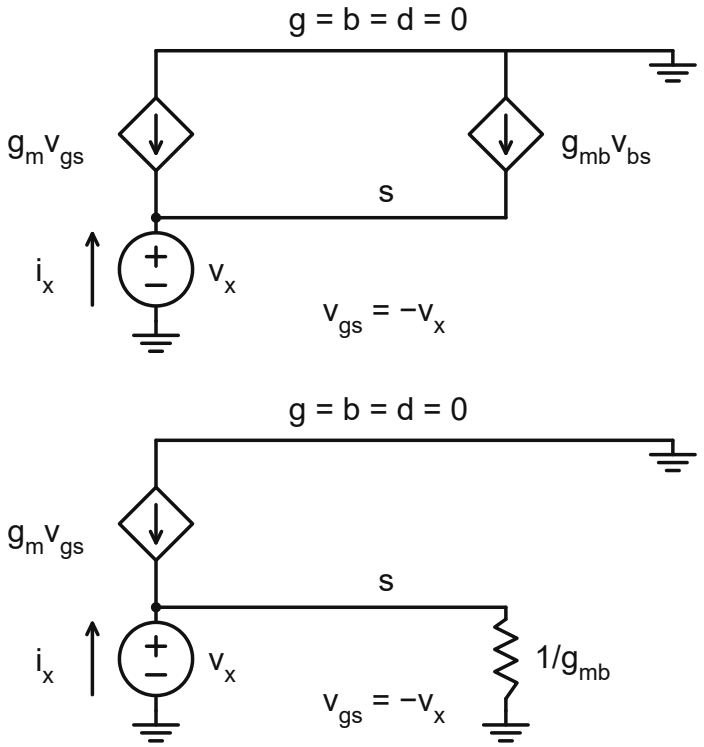
電流源偏壓 Source Follower



$$M_1, M_2 : \text{saturation}, \quad V_{G2} = V_b \ (\text{DC}), \quad I_1 = I_{D2}$$

$$\lambda = 0 : \quad A_v = \frac{g_m}{g_m + g_{mb}}$$

源極測試源與 R_{out}

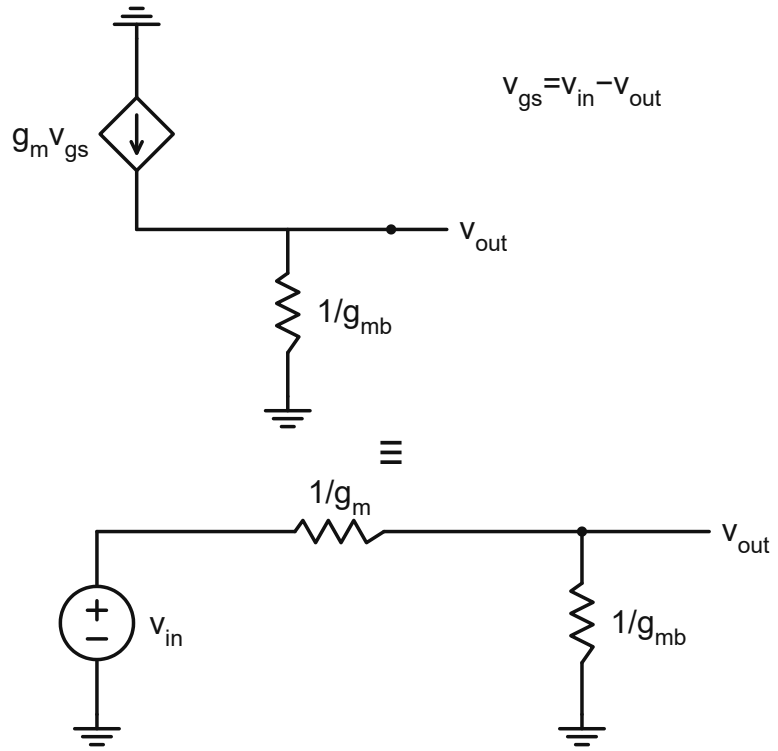


$$\lambda = 0, \quad v_{in} = 0, \quad v_1 = v_{gs} = v_{bs} = -v_x$$

$$i_x + g_m v_1 + g_{mb} v_{bs} = 0 \Rightarrow i_x - g_m v_x - g_{mb} v_x = 0$$

$$i_x = (g_m + g_{mb}) v_x \Rightarrow \boxed{R_{out} = \frac{v_x}{i_x} = \frac{1}{g_m + g_{mb}} = \frac{1}{g_m} \parallel \frac{1}{g_{mb}}}$$

體效應與戴維寧等效

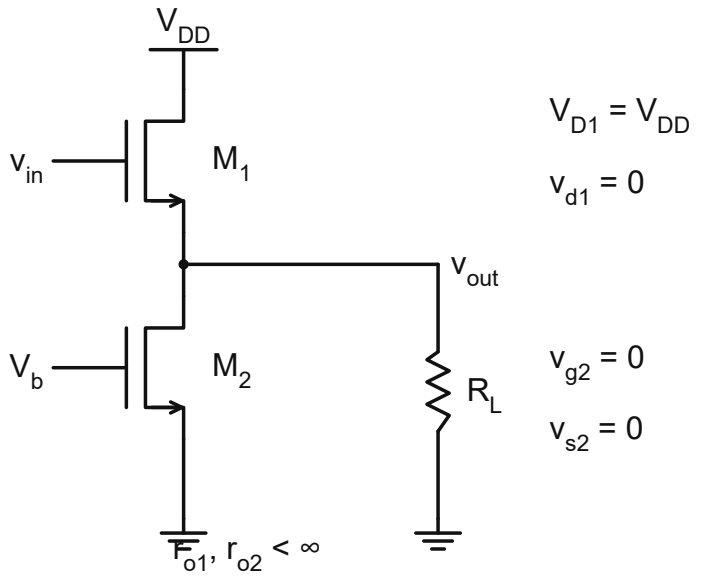


$$\Delta v_{out} = -\Delta V \Rightarrow \Delta v_{GS} = +\Delta V, \quad \Delta V_{TH} = -\eta \Delta V$$

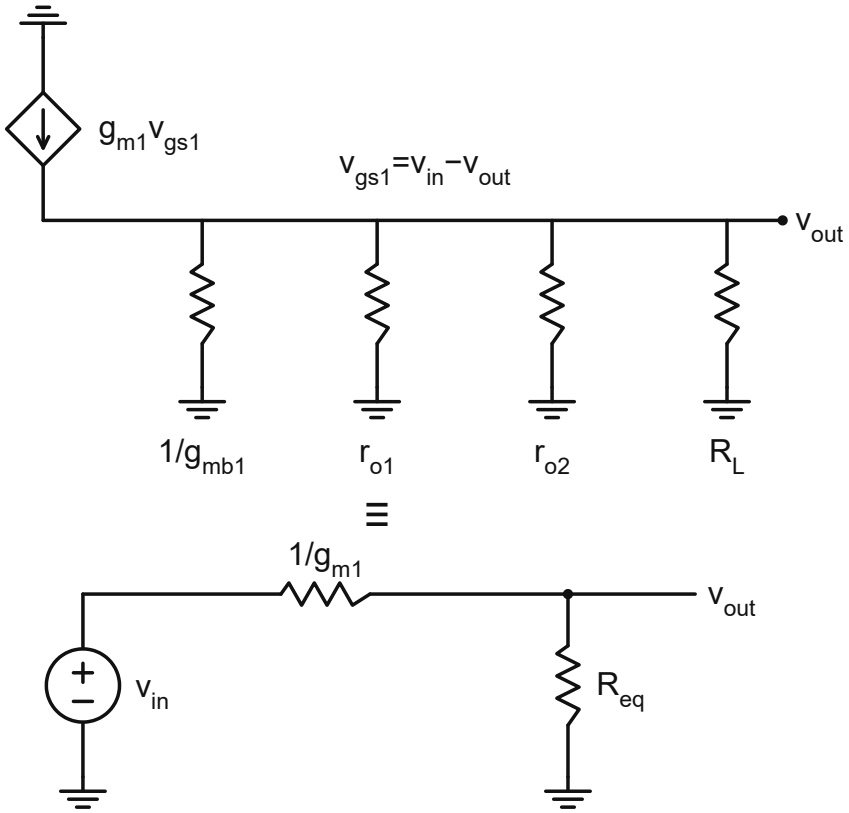
$$\Delta(V_{GS} - V_{TH}) = (1+\eta)\Delta V \Rightarrow \Delta I_D = (g_m + g_{mb})\Delta V$$

$$v_{out} = v_{in}\frac{1/g_{mb}}{1/g_m + 1/g_{mb}} \Rightarrow \boxed{A_v = \frac{g_m}{g_m + g_{mb}}}$$

MOS 電流源與負載電路



有限 r_{o1}, r_{o2} 與負載 R_L



$$R_{eq} = \frac{1}{g_{mb1}} \parallel r_{o1} \parallel r_{o2} \parallel R_L$$

$$\boxed{v_{out} = v_{in}\frac{R_{eq}}{1/g_{m1} + R_{eq}}}$$

有限 r_{o1}, r_{o2} 與負載 R_L（續）

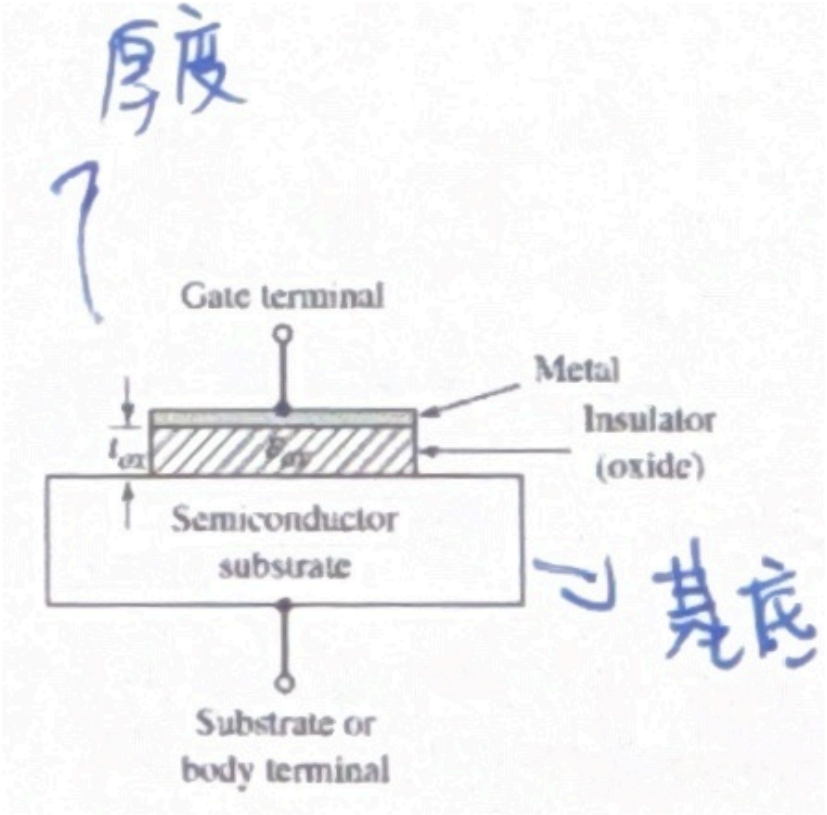
$$\boxed{A_v = \frac{g_{m1}}{g_{m1} + g_{mb1} + 1/r_{o1} + 1/r_{o2} + 1/R_L}}$$

p.04、p.06 — 電容、電流與延遲

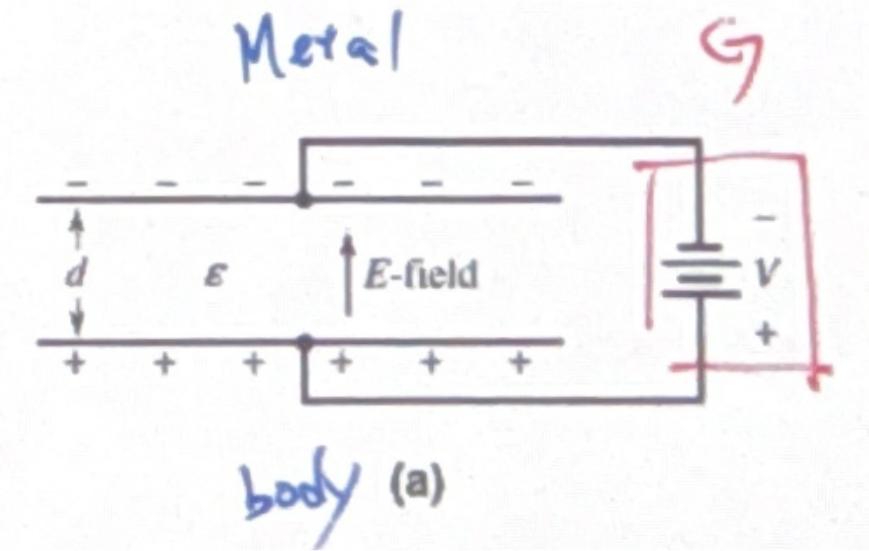
$$I = C\frac{\Delta V}{\Delta t} \quad \Longrightarrow \quad \Delta t = \frac{C}{I}\Delta V$$

$$C = \varepsilon\frac{A}{d}, \qquad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}, \qquad C_g = C_{ox}WL$$

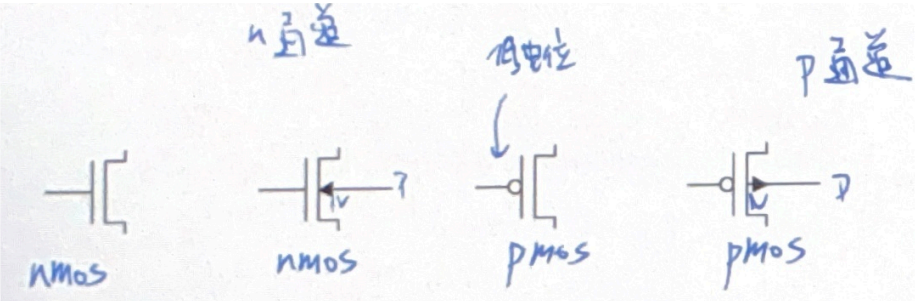
$$\varepsilon_{ox} = 3.9\varepsilon_0, \qquad Q = CV$$



p.04、p.06 — 電容、電流與延遲（續）



p.04 — nMOS / pMOS 符號



p.05–p.06 — 端點電壓、MOS 電容與工作區

$$V_{GS} = V_G - V_S$$
$$V_{GD} = V_G - V_D$$
$$V_{DS} = V_D - V_S = V_{GS} - V_{GD} \geq 0$$

$$\begin{aligned} V_G < 0 &: \text{ Accumulation} \\ 0 < V_G < V_T &: \text{ Depletion} \qquad \text{(p-type body)} \\ V_G > V_T &: \text{ Inversion} \end{aligned}$$

$$\frac{1}{C_{MOS}} = \frac{1}{C_{ox,\text{total}}} + \frac{1}{C_{dep}}$$

$$V_{GS} < V_T : \ I_{DS} \simeq 0, \qquad V_{DS,\text{sat}} = V_{GS} - V_T$$

p.06–p.08 — 通道電荷與載子速度

$$V_C = V_S + \frac{V_{DS}}{2}$$
$$V_{GC} = V_G - V_C = V_{GS} - \frac{V_{DS}}{2}$$
$$\begin{aligned} Q_{\text{channel}} &= C_g(V_{GC} - V_T) \\ &= C_{ox} W L \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right) \end{aligned}$$

$$E = \frac{V_{DS}}{L}, \qquad v = \mu_n E, \qquad t = \frac{L}{v}$$

$$\mu_n \simeq 500\text{–}700 \ \text{cm}^2/(\text{V} \cdot \text{s}) \quad \text{(low electric fields)}$$

p.07–p.09 — 線性區電流推導

$$\begin{aligned} I_{DS} = \frac{Q_{\text{channel}}}{t} &= \frac{C_g(V_{GC} - V_T)}{L/v} \\ &= C_{ox} W \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right) \mu_n E \\ &= \mu_n C_{ox} \frac{W}{L} \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right) V_{DS} \\ &= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[2(V_{GS} - V_T) V_{DS} - V_{DS}^2 \right] \end{aligned}$$

$$\beta = \mu_n C_{ox} \frac{W}{L}, \qquad K_n = \frac{\beta}{2}$$

p.08–p.09 — Shockley 模型

$$V_{DS,\text{sat}} = V_{GS} - V_T = V_{OV}$$

$$I_{DS} = \begin{cases} 0, & V_{GS} \leq V_T \\ \beta \left[V_{OV} V_{DS} - \frac{V_{DS}^2}{2} \right], & V_{GS} > V_T, \ 0 \leq V_{DS} < V_{OV} \\ \frac{\beta}{2} V_{OV}^2, & V_{GS} > V_T, \ V_{DS} \geq V_{OV} \end{cases}$$

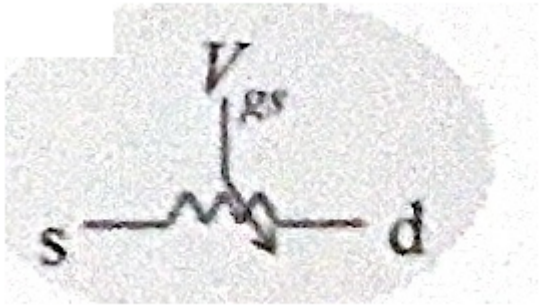
$$\begin{aligned} I_{DS,\text{sat}} &= \beta \left[(V_{GS} - V_T) V_{OV} - \frac{V_{OV}^2}{2} \right] \\ &= \frac{\beta}{2} (V_{GS} - V_T)^2 \end{aligned}$$

p.08–p.09 — 深線性區電阻

$$V_{DS} \ll 2(V_{GS} - V_T)$$

$$I_{DS} \simeq \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)V_{DS}$$

$$\boxed{R_{on} = \frac{V_{DS}}{I_{DS}} \simeq \frac{1}{\mu_n C_{ox}(W/L)(V_{GS} - V_T)}}$$



p.11、p.17 — 通道長度調變與 r_o

$$i_D = K_n(v_{GS} - V_{TN})^2(1 + \lambda v_{DS}) \tag{4.16}$$

$$-V_A = -\frac{1}{\lambda}, \qquad \text{slope} = \frac{1}{r_o}$$

$$r_o = \left(\frac{\partial i_D}{\partial v_{DS}}\bigg|_{v_{GS}=V_{GSQ}} \right)^{-1} \tag{4.17}$$

$$\begin{aligned} r_o &= \left[\lambda K_n (V_{GSQ} - V_{TN})^2\right]^{-1} \\ &\simeq [\lambda I_{DQ}]^{-1} \end{aligned} \tag{4.18}$$

p.12 — 直流偏壓與負載線

$$V_{DD} = 5\ \text{V},\ R_1 = 30\ \text{k}\Omega,\ R_2 = 20\ \text{k}\Omega$$

$$R_D = 20\ \text{k}\Omega, \quad I_D = 0.1\ \text{mA}$$

$$V_G = \frac{R_2}{R_1 + R_2} V_{DD} = \frac{20}{50}(5) = 2\ \text{V}$$

$$V_{GS} = 2\ \text{V}, \qquad V_{DS} = 5 - (0.1)(20) = 3\ \text{V}$$

$$V_{DS} = V_{DD} - I_D R_D, \quad I_{D,\max} = \frac{V_{DD}}{R_D}, \quad \text{slope} = -\frac{1}{R_D}$$

p.16 — 大訊號、偏壓與小訊號分離

$$v_{GS} = V_{GSQ} + v_i = V_{GSQ} + v_{gs} \tag{4.1}$$

$$i_D = K_n(v_{GS} - V_{TN})^2 \tag{4.2}$$

$$\begin{aligned} i_D &= K_n[V_{GSQ} + v_{gs} - V_{TN}]^2 \\ &= K_n[(V_{GSQ} - V_{TN}) + v_{gs}]^2 \end{aligned} \tag{4.3a}$$

$$\begin{aligned} i_D = {} & K_n(V_{GSQ} - V_{TN})^2 \\ & + 2K_n(V_{GSQ} - V_{TN})v_{gs} + K_n v_{gs}^2 \end{aligned} \tag{4.3b}$$

$$|v_{gs}| \ll 2(V_{GSQ} - V_{TN}) \tag{4.4}$$

$$i_D = I_{DQ} + i_d \tag{4.5}$$

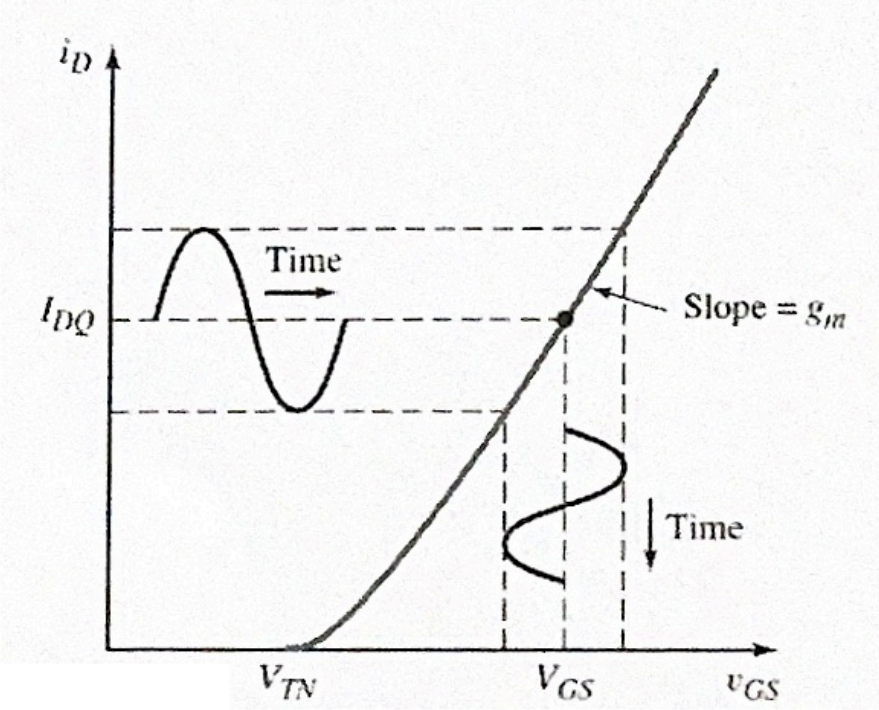
p.16 — 轉導 g_m

$$i_d = 2K_n(V_{GSQ} - V_{TN})v_{gs} \tag{4.6}$$

$$g_m = \frac{i_d}{v_{gs}} = 2K_n(V_{GSQ} - V_{TN}) \tag{4.7}$$

$$g_m = \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{GS}=V_{GSQ}} = 2K_n(V_{GSQ} - V_{TN}) \tag{4.8a}$$

$$g_m = 2\sqrt{K_n I_{DQ}} \tag{4.8b}$$



p.17 — 共源級輸出與交流關係

$$v_{DS} = v_O = V_{DD} - i_D R_D \tag{4.9}$$

$$\begin{aligned} v_O &= V_{DD} - (I_{DQ} + i_d) R_D \\ &= (V_{DD} - I_{DQ} R_D) - i_d R_D \end{aligned} \tag{4.10}$$

$$v_o = v_{ds} = -i_d R_D \tag{4.11}$$

$$i_d = g_m v_{gs} \tag{4.12}$$

$$v_{gs} = v_i \quad \text{or} \quad V_{gs} = V_i \tag{4.13}$$

$$i_d = g_m v_{gs} \quad \text{or} \quad I_d = g_m V_{gs} \tag{4.14}$$

$$v_{ds} = -i_d R_D \quad \text{or} \quad V_{ds} = -I_d R_D \tag{4.15}$$

p.17–p.18 — 體效應與臨界電壓

$$i_D = K_n (v_{GS} - V_{TN})^2 \tag{4.22}$$

$$V_{TN} = V_{TN0} + \gamma \left[\sqrt{2\phi_f + v_{SB}} - \sqrt{2\phi_f} \right] \tag{4.23}$$

$$v_{SB} \geq 0, \qquad v_{bs} = -v_{sb}$$

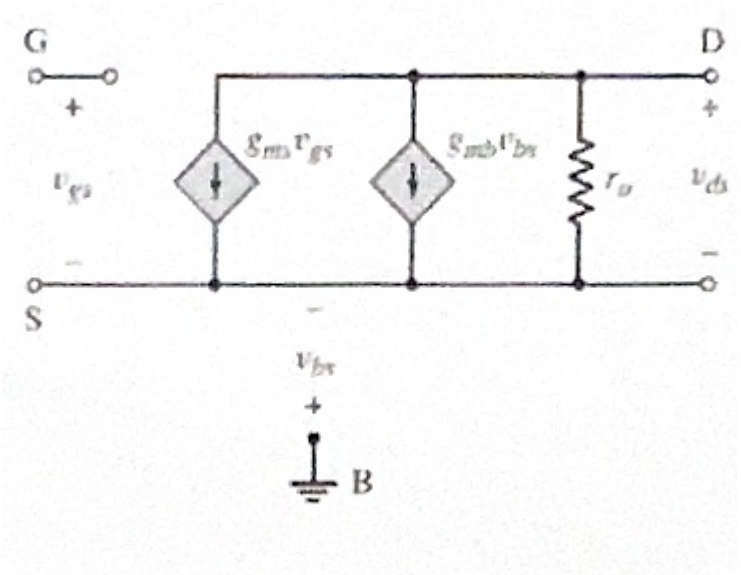
$$\begin{aligned} g_{mb} &= \left. \frac{\partial i_D}{\partial v_{BS}} \right|_Q = - \left. \frac{\partial i_D}{\partial v_{SB}} \right|_Q \\ &= - \left(\frac{\partial i_D}{\partial V_{TN}} \frac{\partial V_{TN}}{\partial v_{SB}} \right) \Bigg|_Q \end{aligned} \tag{4.24}$$

p.18 — 體轉導 g_{mb}

$$\frac{\partial i_D}{\partial V_{TN}} = -2K_n (v_{GS} - V_{TN}) = -g_m \tag{4.25a}$$

$$\frac{\partial V_{TN}}{\partial v_{SB}} = \frac{\gamma}{2\sqrt{2\phi_f + v_{SB}}} = \eta \tag{4.25b}$$

$$\boxed{g_{mb} = -(-g_m)\eta = g_m \eta} \tag{4.26}$$



p.18 — η 數值與代入

$$\phi_f = 0.35\ \text{V}, \quad \gamma = 0.35\ \text{V}^{1/2}, \quad v_{SB} \geq 0$$

講義標值：$\eta \simeq 0.23, \quad 0 \leq \eta \leq 0.23$

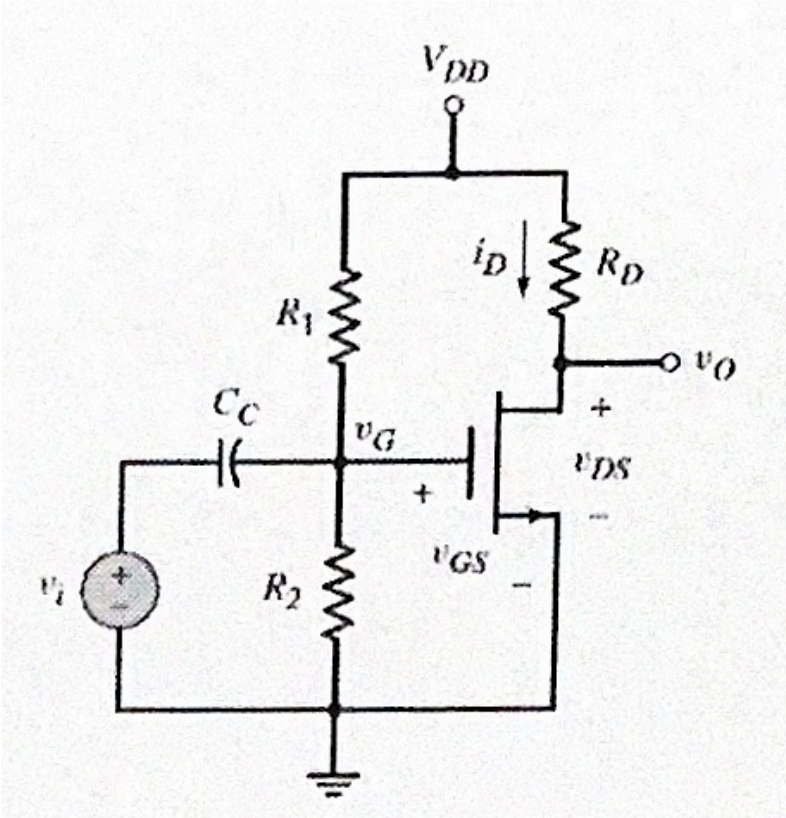
公式代入：

$$\eta_{\max} = \frac{\gamma}{2\sqrt{2\phi_f}}$$

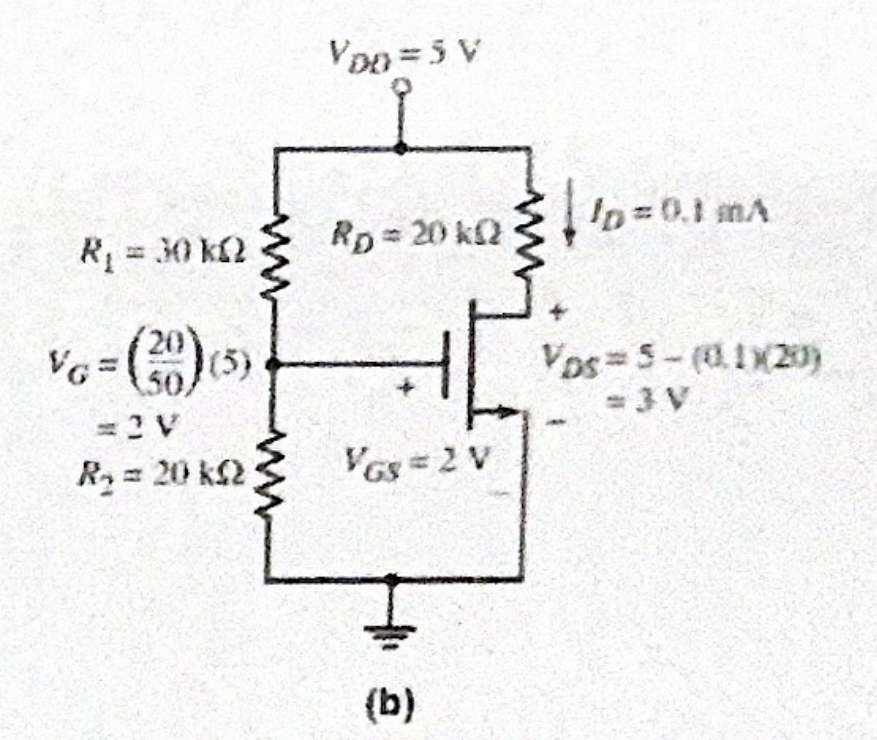
$$= \frac{0.35}{2\sqrt{0.70}} \simeq 0.209$$

$$0 \leq \eta \leq 0.209$$

p.12 · Fig.3.24–3.25：分壓偏壓 CS

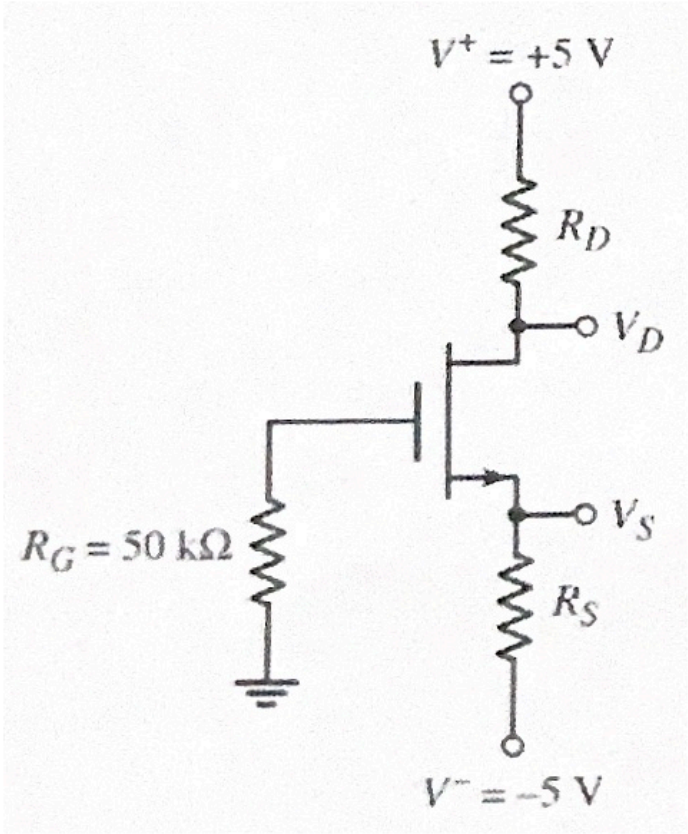


p.12 · Fig.3.24–3.25：分壓偏壓 CS（續）

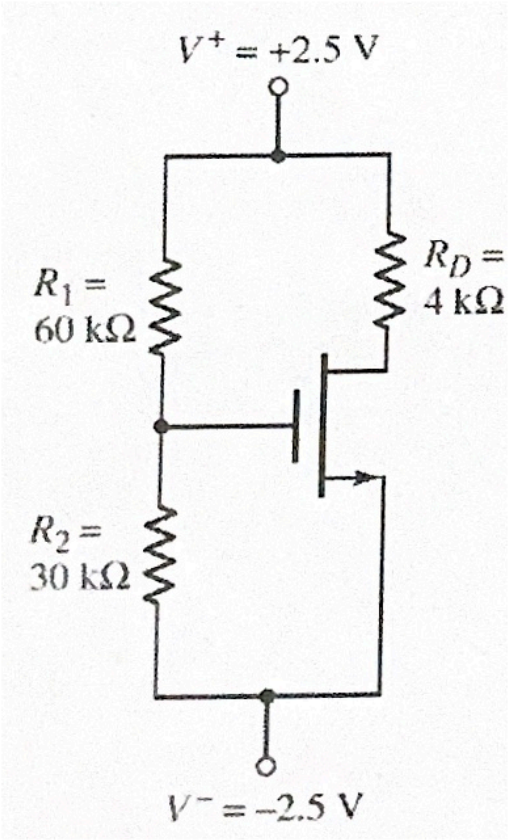


VDD	R_1	R_2	RD
5 V	30 kΩ	20 kΩ	20 kΩ
ID = 0.1 mA	VGS = 2 V	VDS = 3 V	IG = 0

p.12 · Fig.3.27：雙電源源極偏壓



p.12–13 · Fig.3.28、3.33：分壓與源極電阻



p.12–13 · Fig.3.28、3.33：分壓與源極電阻（續）

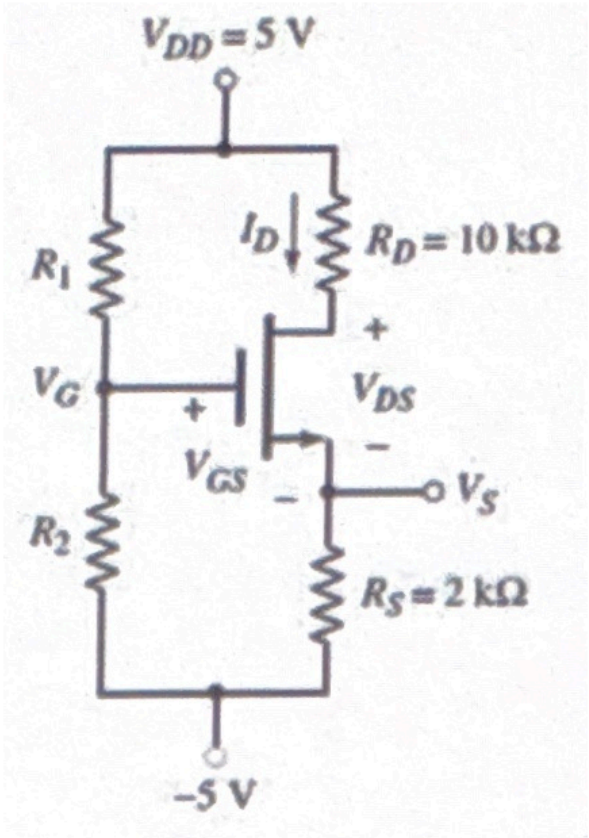


Fig.	V⁺ / V⁻	R₁ / R₂	RD / RS
3.28	+2.5 / −2.5 V	60 / 30 kΩ	4 kΩ / 0
3.33	+5 / −5 V	R₁ / R₂	10 / 2 kΩ

p.13 · Fig.3.32；p.18 · Fig.4.21：電流偏壓 CS

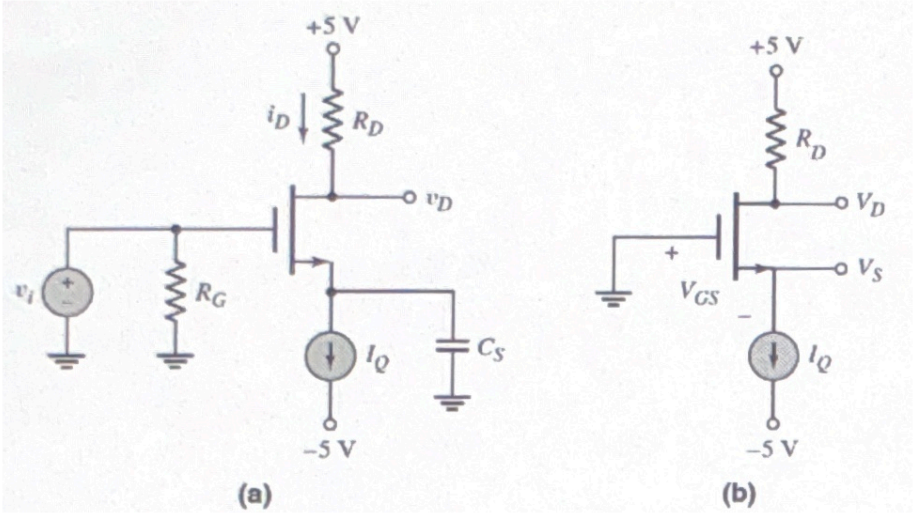
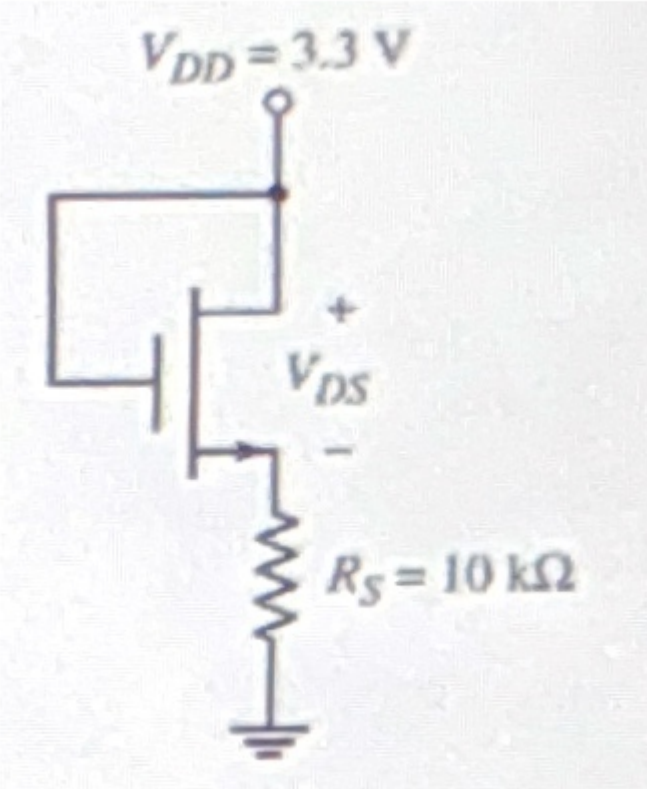


Fig.3.32	RD、RG、IQ、CS
Fig.4.21	RD = 7 kΩ；RG = 200 kΩ；IQ = 0.5 mA

p.13–14 · Fig.3.43–3.44：MOS 自偏壓



p.13–14 · Fig.3.43–3.44：MOS 自偏壓（續）

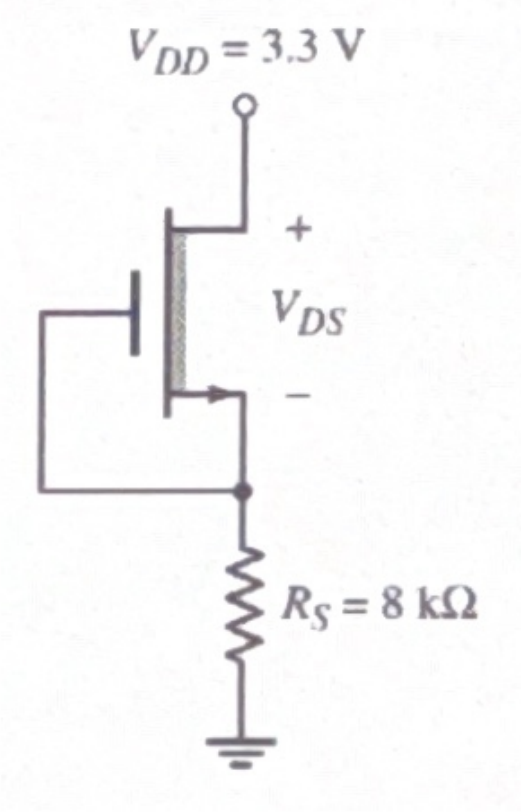
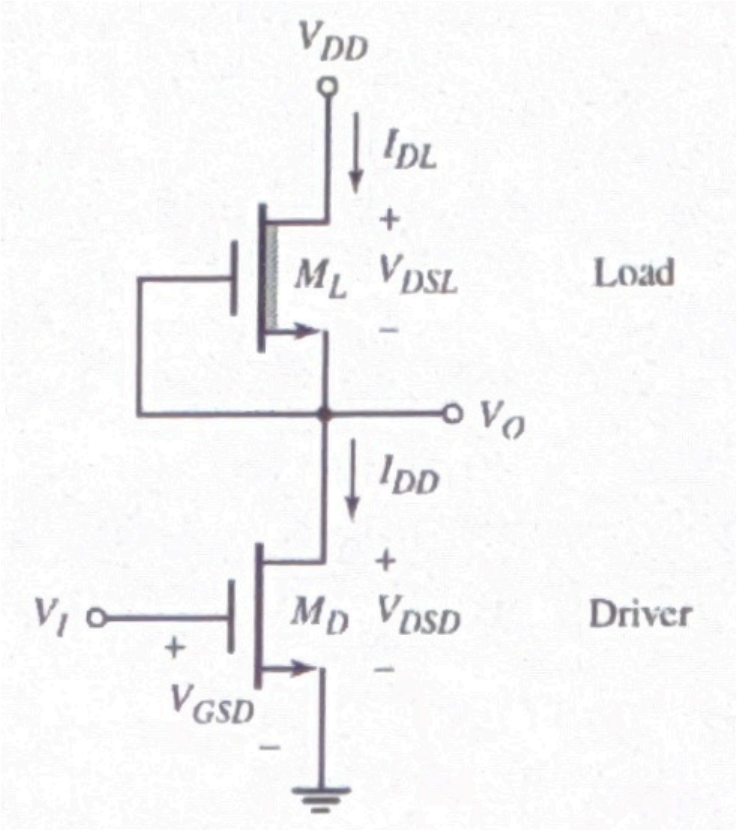
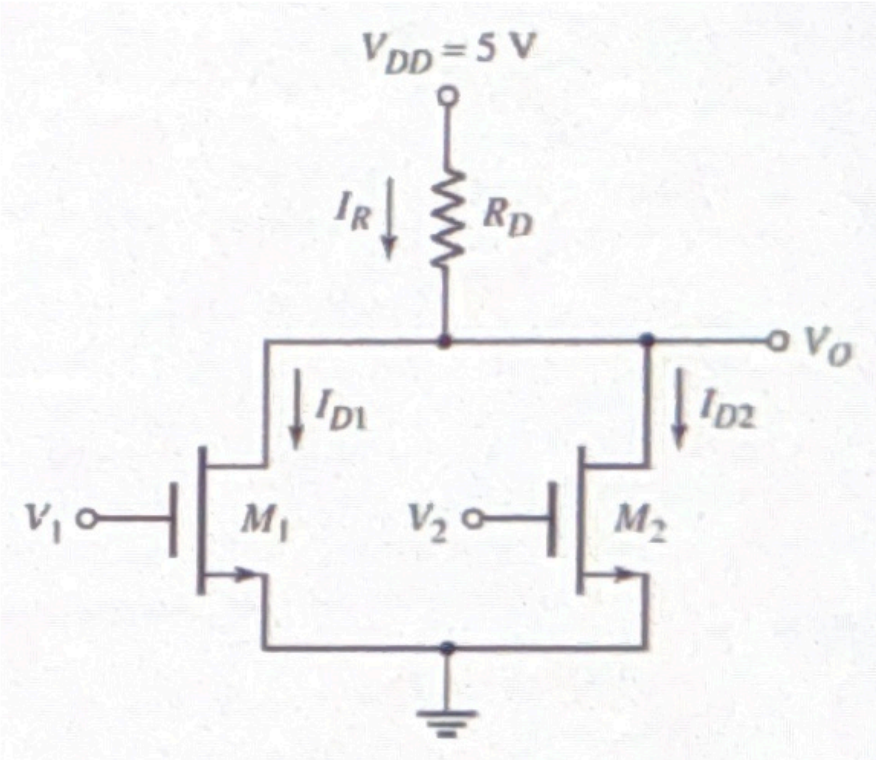


Fig.	VDD	RS	接法
3.43	3.3 V	10 kΩ	Gate = Drain
3.44	3.3 V	8 kΩ	Gate = Source；Depletion

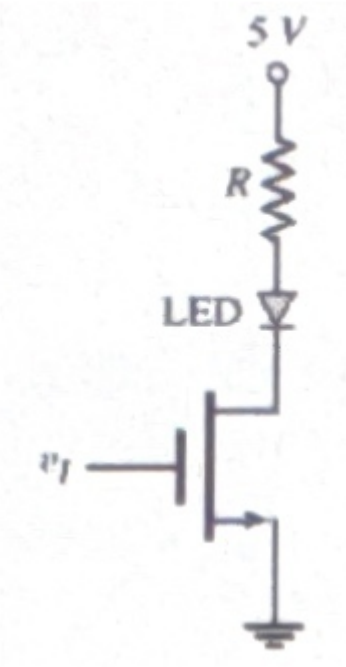
p.14 · Fig.3.38–3.39：耗盡型負載反相器



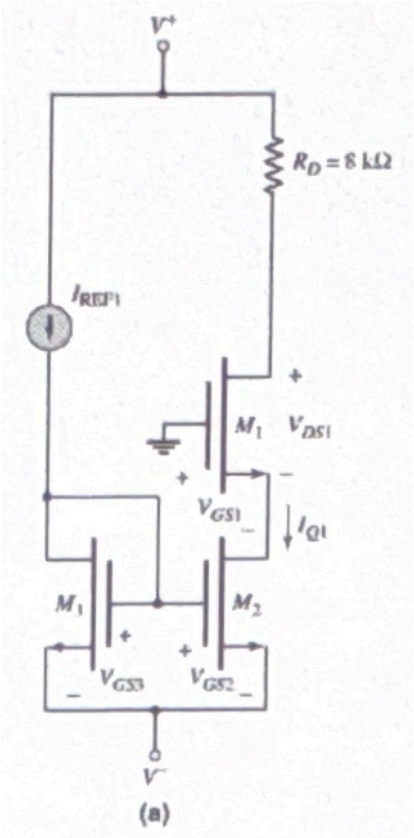
p.14 · Fig.3.46：雙輸入 NMOS NOR



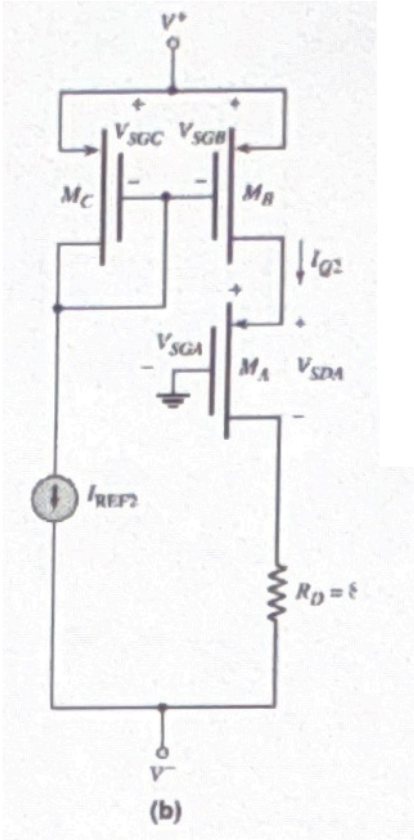
p.15 · Fig.3.48：LED 驅動



p.15 · Fig.3.49(a)：NMOS 電流鏡

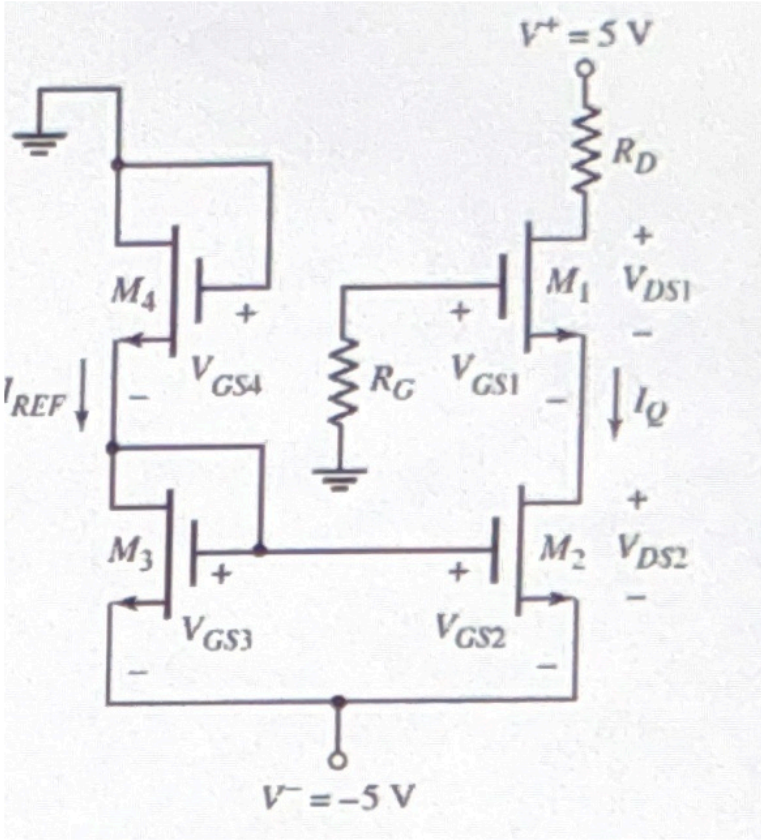


p.15 · Fig.3.49(b)：PMOS 電流鏡

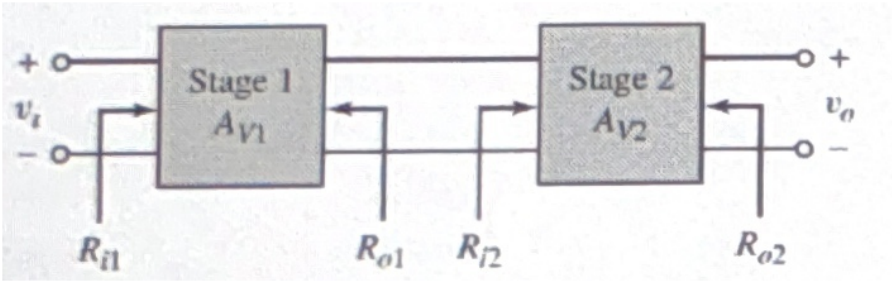


R_D 標值：原圖截斷

p.15 · Fig.3.50：MOSFET 定電流源



p.15 · Fig.3.51：兩級放大器



p.15、21 · Fig.3.52 / 4.49：CS → Source Follower

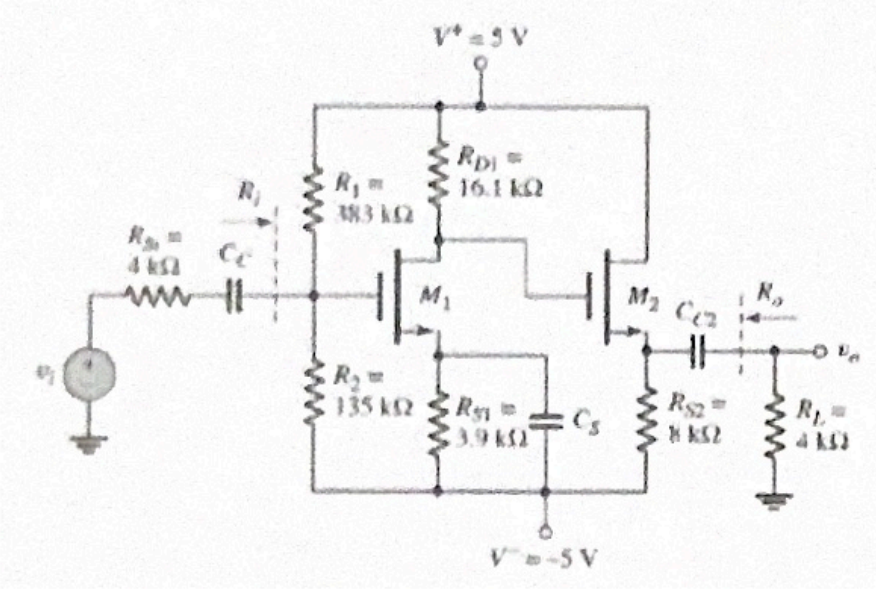
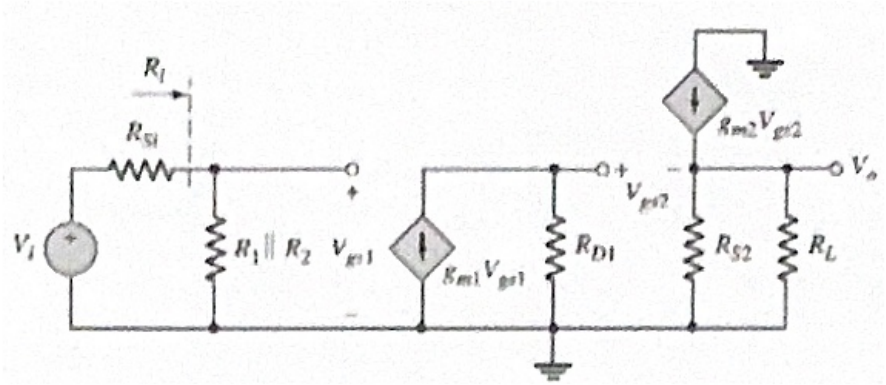


Fig.4.49	RSi = 4 kΩ	RD1 = 16.1 kΩ
R₁（原圖模糊）	R₂ = 135 kΩ	RS1 = 3.9 kΩ
RS2 = 8 kΩ	RL = 4 kΩ	V⁺/V⁻ = ±5 V
Fig.3.52	同拓撲；RL = 4 kΩ	其餘阻值以符號標示

p.21 · Fig.4.50：CS → SF 小訊號模型



p.15、21 · Fig.3.53 / 4.51：Cascode

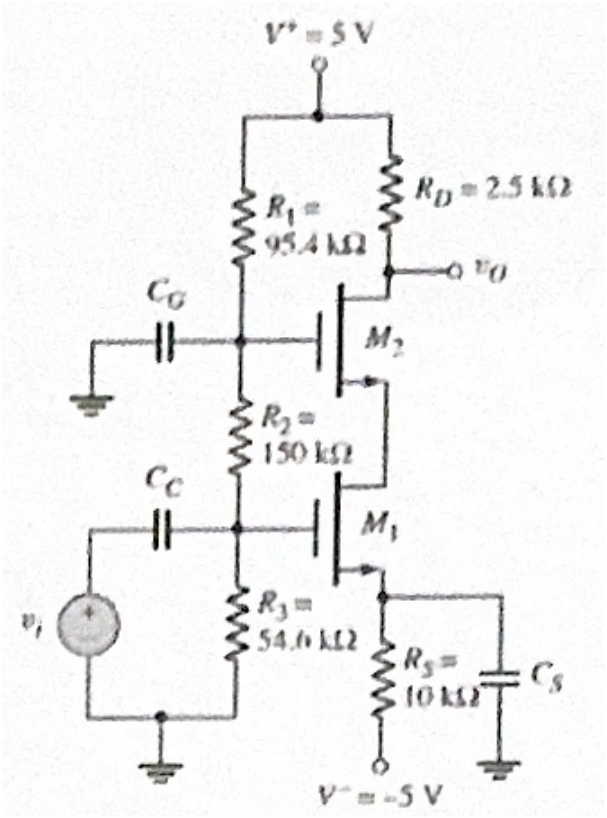
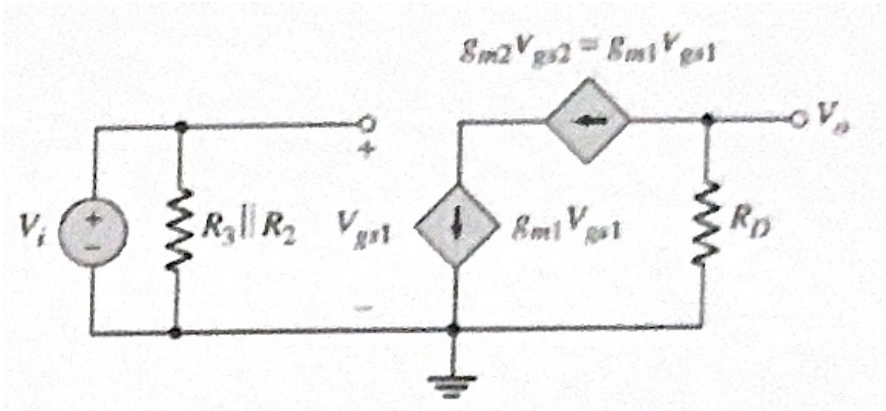
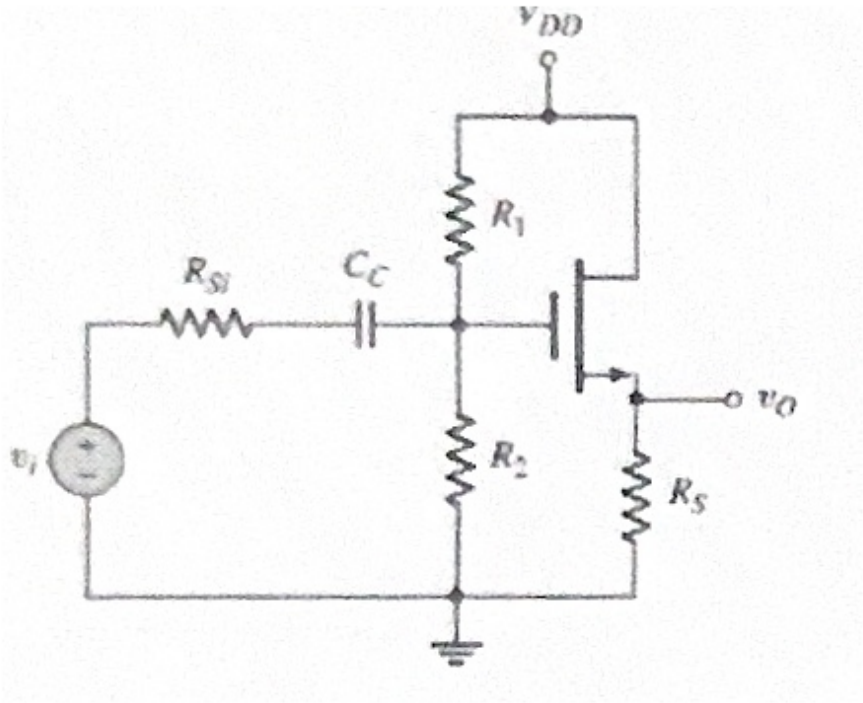


Fig.4.51	R_1 = 95.4 kΩ	R_2 = 150 kΩ
R_3 = 54.6 kΩ	RD = 2.5 kΩ	RS = 10 kΩ
Fig.3.53	同拓撲；V^+/V^- = ±5 V	電阻以符號標示

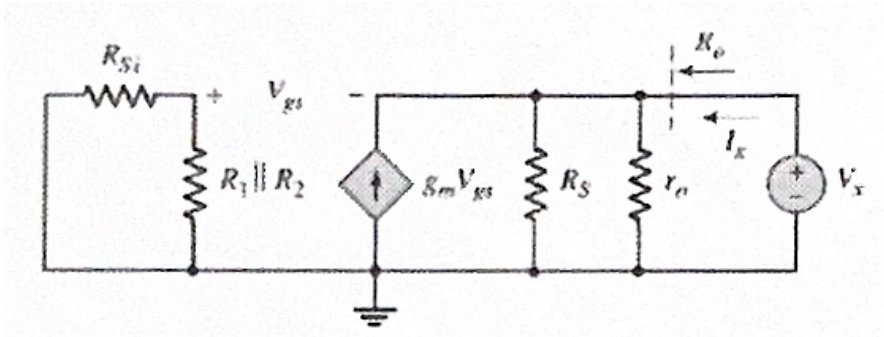
p.21 · Fig.4.52：Cascode 小訊號模型



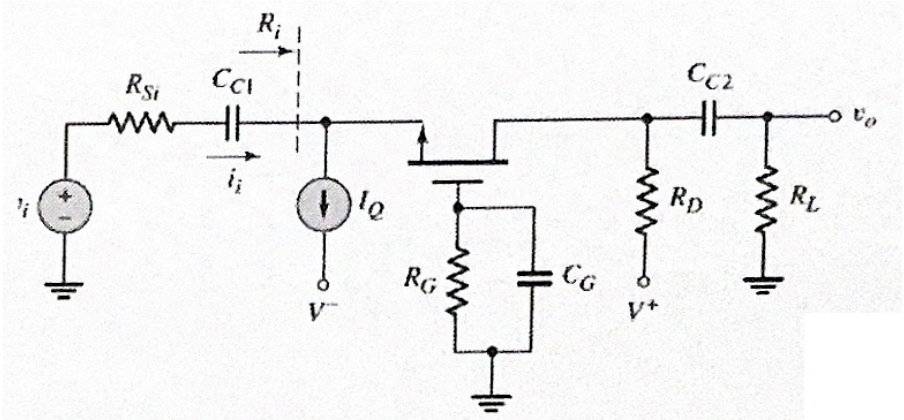
p.19 · Fig.4.26：分壓偏壓 Source Follower



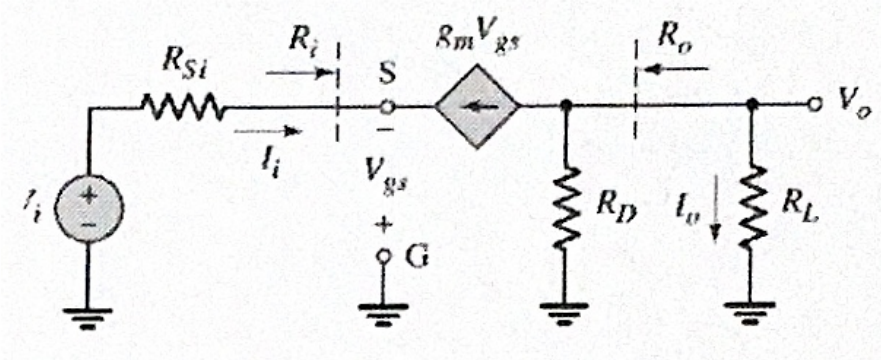
p.19 · Fig.4.30：SF 輸出測試與偏壓網路



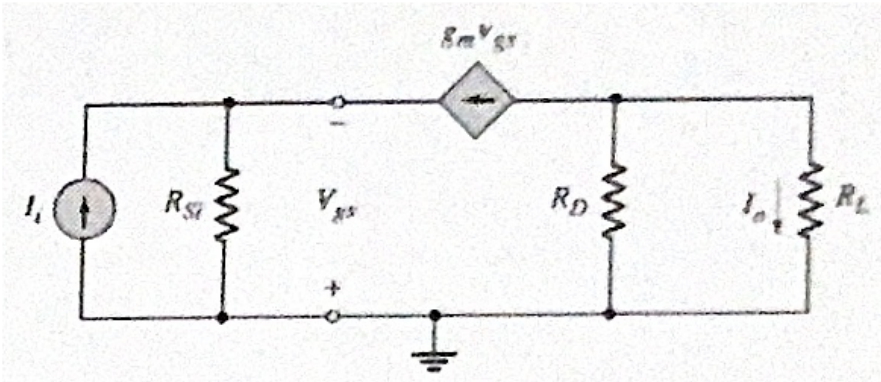
p.19 · Fig.4.32：Common Gate 偏壓與耦合



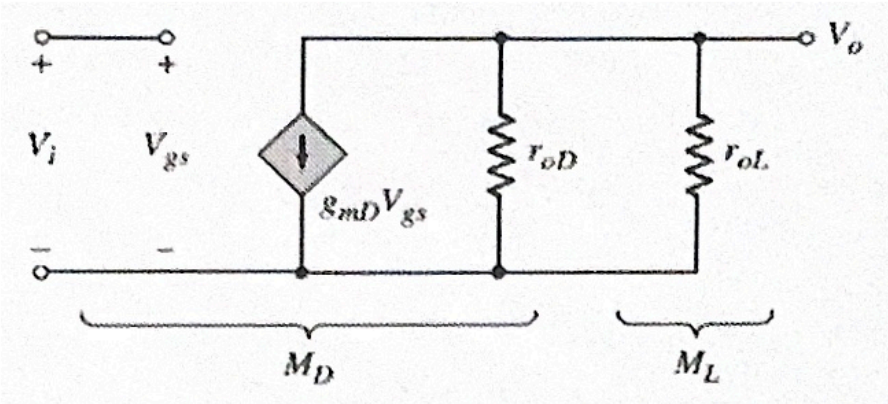
p.19 · Fig.4.33：Common Gate 小訊號模型



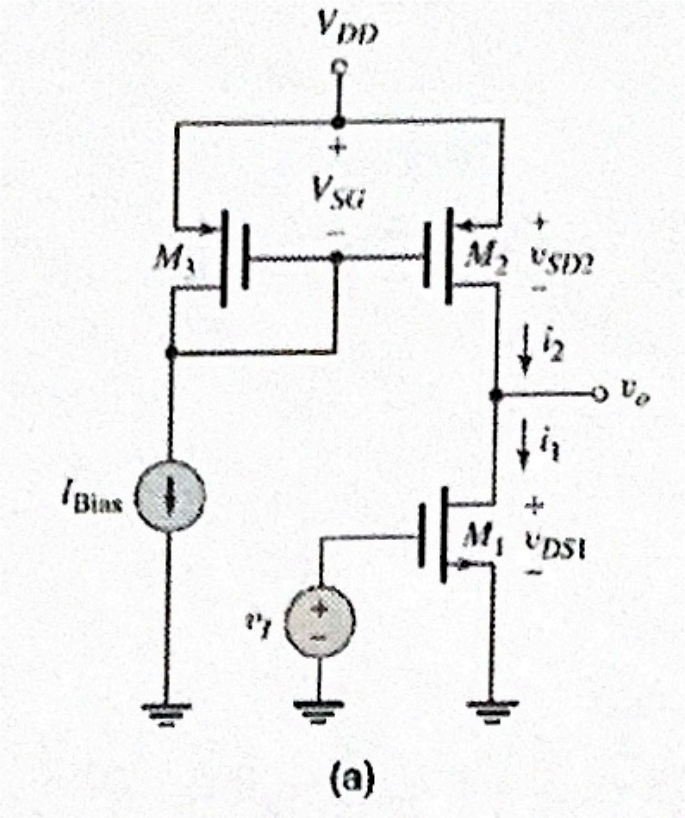
p.19 · Fig.4.34：Common Gate Norton 模型



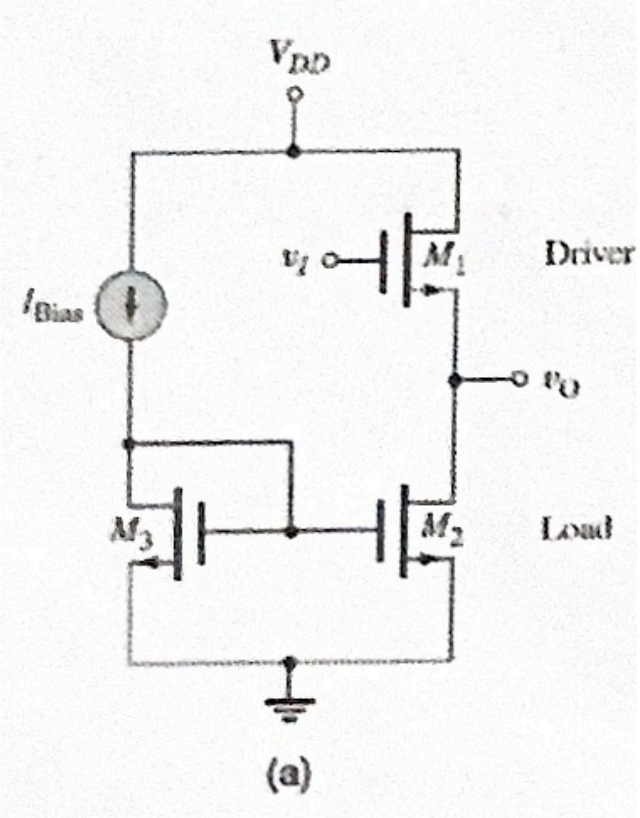
p.20 · Fig.4.44：耗盡負載小訊號模型



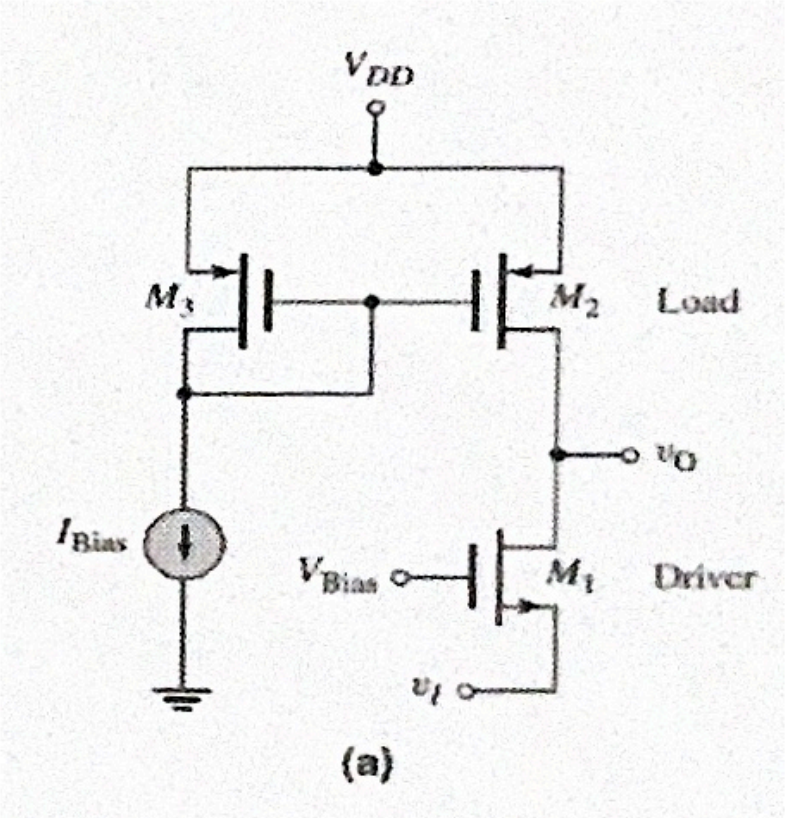
p.20 · Fig.4.45：PMOS 鏡負載共源級



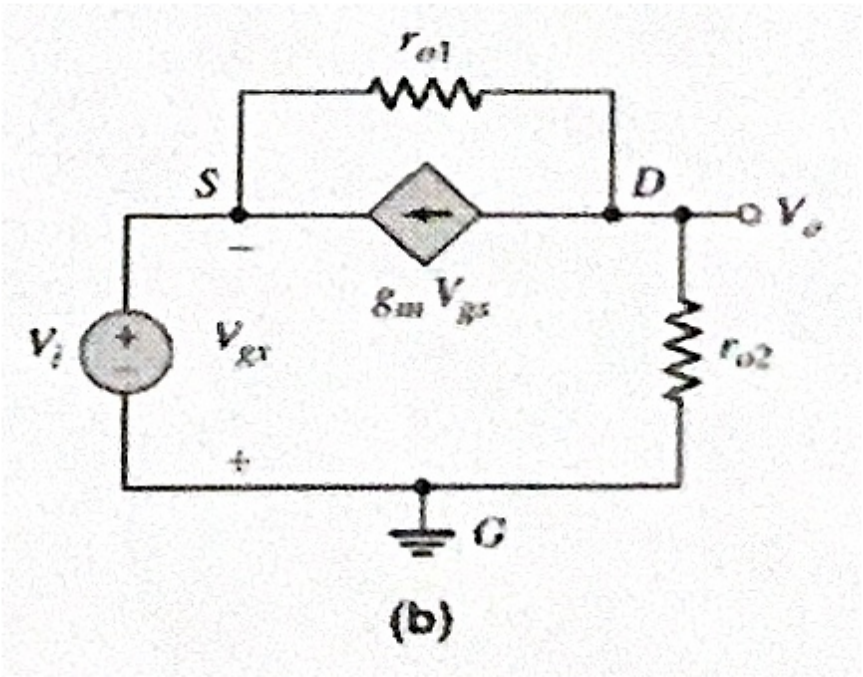
p.20 · Fig.4.47(a)：鏡偏壓 Source Follower



p.20 · Fig.4.48(a)：CMOS Common Gate



p.20 · Fig.4.48(b)：CMOS CG 小訊號模型



p.20 · Fig.4.48(c)：CMOS CG 輸出測試

